1 About this Open Learning Module

Target Audience and Prerequisites

This Open Learning module is aimed at new recruits in Great Britain into the role of Signalling Technician on either the installation or maintenance sides. It can also be used to update or refresh the knowledge of experienced technicians who are undergoing re-certification of their competence.

Apart from a basic Secondary Education, an ability to understand moderately complex English language and the presentation of information in the form of diagrams, plus a reasonable curiosity and a willingness to learn, the only requirement is that you should have successfully completed the study of

- the associated Open Learning Module “Introduction to Railway Signalling”
- Signalling For Technicians Module 1 “Circuit Reading”

or have acquired equivalent basic knowledge about the purpose and implementation of railway signalling, the forms of diagrams and schematic drawings used and the various symbols and methods of labelling employed.

Objectives

This module is intended to provide an introduction to and an appreciation of the operation and use of

- mechanical signals
- colour light signals
- route/junction indicators
- the automatic warning system
- train protection warning system

What you will require

Apart from this Open Learning manual the only other material you will require is your Network Rail Rule Book, the Track Safety Handbook and the Signalling Maintenance Specifications.

The Results expected from this Module

When you have completed this module, the assessment and been awarded your certificate, you should

- have an appreciation of the purpose of railway signalling
- be able to identify and understand the construction and use of the following types of mechanical signals

  - stop

  - distant

- junction
- subsidiary
- shunting

- be able to identify and understand the construction and use of the following types of colour light signals
  - single aspect
  - two aspect
  - three aspect
  - four aspect
  - single lamp searchlight
  - two lamp searchlight
- be able to identify and understand the construction and use of the following types of route/junction indicator
  - position light (feather) junction indicator
  - multi-lamp (theatre type) route indicator
  - stencil type route indicator
  - fibre optic versions of the theatre and stencil type route indicators

- be able to identify and understand the construction and use of the following types of position light signals

  - three lamp

  - two lamp

- be able to identify and understand the construction and use of the track equipment part of the automatic warning system (AWS) and train protection warning system (TPWS)

- be able to correctly use and understand the jargon commonly used in association with signals, AWS and TPWS.

Jargon

In the main body of this module the officially recognised jargon descriptions are used and some commonly used variations are introduced. The Glossary of Terms (see Section 11) provides some of the other relevant variations which are used.

A Safety Warning

This Open Learning Module is designed to inform you about railway points. However, please remember that, until you have undergone proper training and been certified as competent in Personal Track Safety (PTS) **and you have been authorised by your Supervisor**, you are **forbidden** to go onto any part of the railway that is not open to passengers, unless you are properly supervised and it is an official part of your training.

Study and remember what it says in your Rule Book and the Track Safety Handbook. There is no requirement for you to go onto the railway to successfully complete this course of study. Of course, your supervisor or appointed mentor may use their discretion and properly arrange for you to go onto the railway to look at practical examples of the points, associated equipment and layouts described in this manual and which you have not already had an opportunity to observe in the course of your work and other training.

2 Railway Signalling

2.1 Introduction

The majority of the information concerning the areas examined within this Module can be found in the **Rule Book**. Make sure you have a copy of the rule book available as specific Self-Assessment Questions will require verification from this book.

The basic purpose of the railway signalling system is to

- maintain a safe distance between following trains on the same track
- safeguard the movement of trains at junctions and when crossing a path which could be taken by another train
- regulate the passage of trains according to the service density and speed required

In this module you are going to examine the two basic types of signals used in the rail system, namely

- mechanical signals
- colour light signals

Mechanical signals are operated **manually** by wires incorporating **cranks and wheels**.

The colour light signals are operated by **electricity**.

The automatic warning system used on the railway will also be examined. This system is used with

- mechanical distant signals
- colour light signals capable of displaying cautionary aspects

Not covered in the manual is the latest evolution of the automatic warning system which is Automatic Train Protection. This automatically applies the train brakes if it runs past a red signal. This will be dealt with later in other training courses.

2.2 Section Summary

This section has provided an introduction to the signalling system used on the railway and has outlined the importance of the Rule Book.

Section Three will look at mechanical signals.

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3 Mechanical Signals

3.1 Introduction

There are three types of mechanical signals

- main
- subsidiary
- ground

Each of these will now be examined in greater detail.

3.2 Main Mechanical Signals

There are two types of main mechanical signal

- stop
- distant

Mechanical Signal Post and Fittings

Figure 1 illustrates a mechanical signal complete with post and fittings.

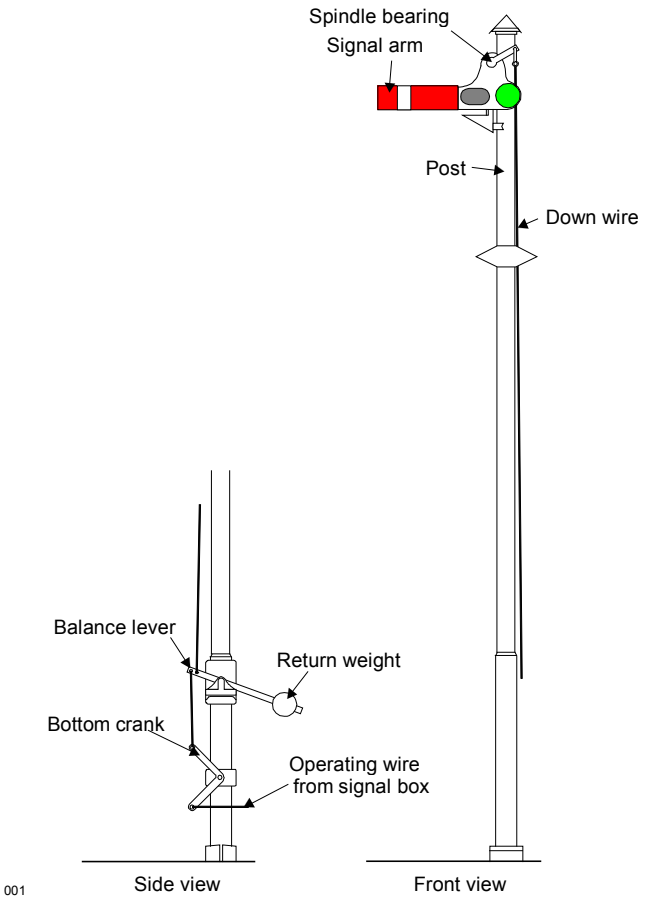


Figure 1

The signal arm fits on a bracket which bolts or clamps to a straight post. The signal arm rotates around the pivot spindle bearing shown in figure 1.

Operation

The signaller pulls a lever which pulls a wire. This in turn pulls the bottom crank which transforms the horizontal pull into a vertical pull. The crank is attached to a balance lever which carries a return weight. The vertical pull actually pulls the signal arm into its off position.

When the signaller returns the lever to the normal position the return weight drops and takes up the slack in the wire run to the signal box. The signal arm returns to the on position under its own weight.

Stop

The stop signal arm is illustrated in figure 2.

Semaphore Stop Signal Arm

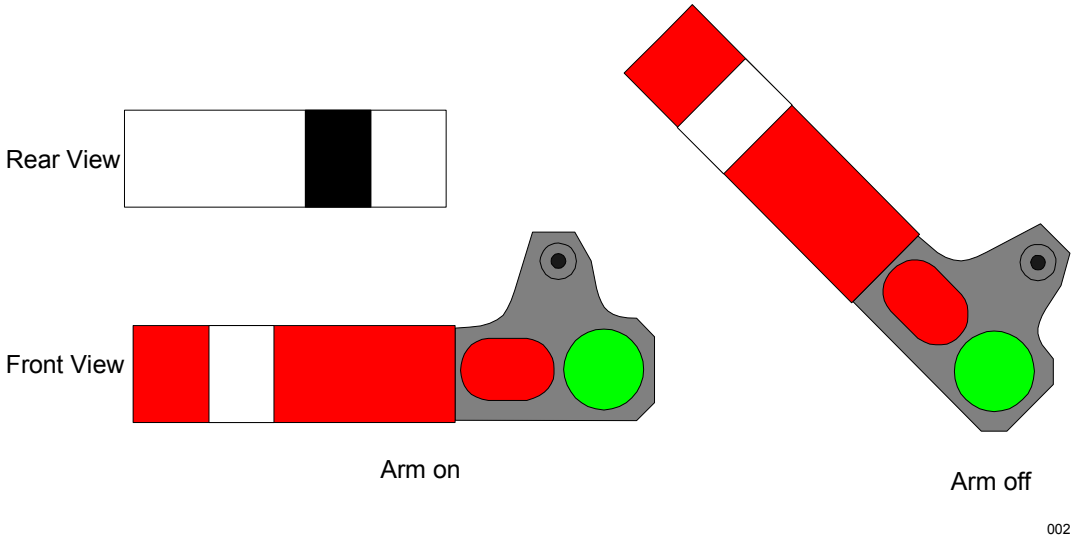


Figure 2

Stop signals consist of a **red** arm with spectacles of **red** and **green**.

In an approximately horizontal position the arm indicates **stop** to the driver. The night-time aspect will be red.

When the signal arm is at approximately 45° it indicates **proceed** to the driver. The night-time aspect will be green.

In the horizontal position the signal is said to be **on**.

At the 45° angle the signal is said to be **off**.

Semaphore Distant Signal Arm

Distant

The distant signal arm is illustrated in figure 3.

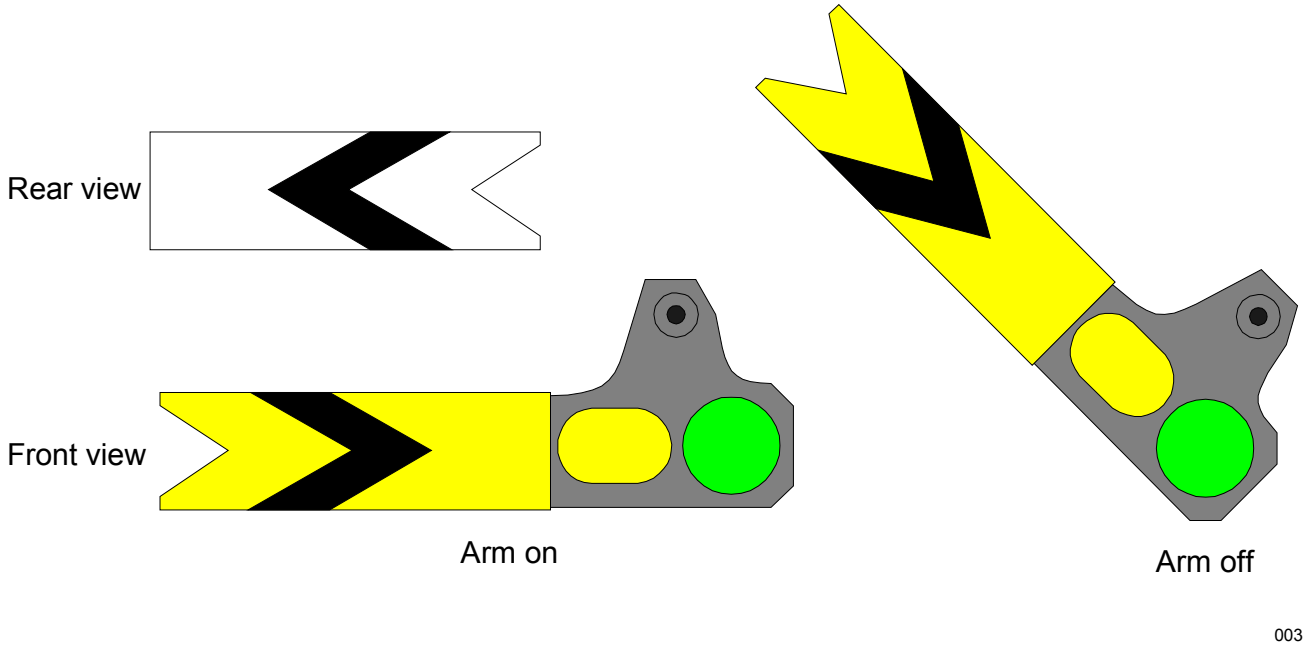


Figure 3

Distant signals consist of a **yellow arm** with spectacles of **yellow** and **green**.

When this signal is horizontal it indicates to the driver to proceed at caution and be prepared to stop at a stop signal. There may be more than one stop signal in advance of a distant signal, controlled from the same signal box. The night-time aspect will be yellow.

When this signal is at an approximately 45° angle it indicates to the driver to proceed at line speed, as all relevant stop signals controlled by this signal box have been cleared. The night-time aspect will be green.

In the horizontal position the signal is said to be **on**.

At the 45° angle the signal is said to be **off.**

Junction Signalling

Semaphore Junction Signals

Figure 4 illustrates a typical example of a mechanical junction signal on part of an overhead gantry.

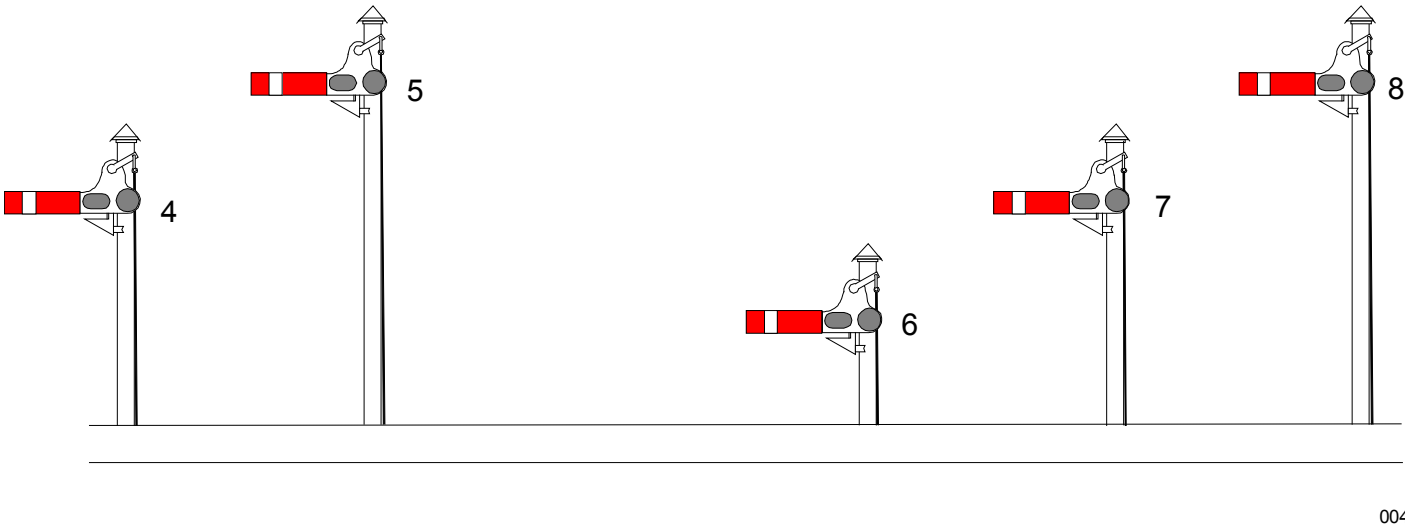


Figure 4

Schematic of Junction

Figure 5 shows the schematic version of the signal illustrated in figure 4 and its associated layout.

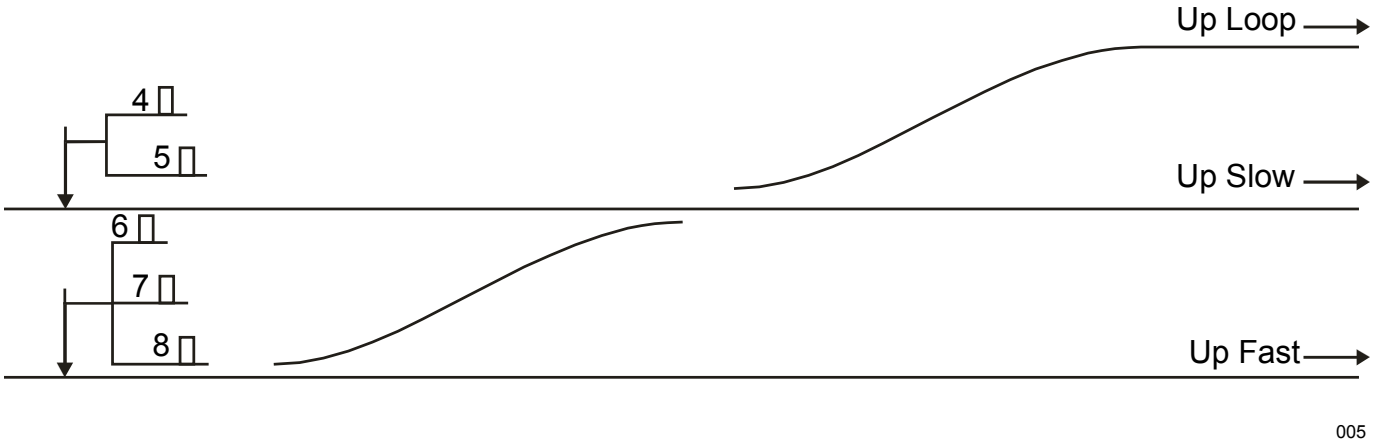


Figure 5

In this junction signalling there are two arrays of signals

- one group of two (signals 4 and 5)
- one group of three (6, 7 and 8)

Signals 4 and 5 are associated with the up slow line.

6, 7 and 8 are associated with the up fast line.

On the gantry signals associated with a different line are usually separated by a wide space.

Signals associated with the same line are grouped closer together. This is illustrated in figure 4.

Consider signal 8 - this is the tallest signal in its group of three.

When the signal is off this indicates to the driver that he is being signalled for the main up fast route.

The main route, usually fastest, is indicated by the tallest signal.

Signal 7 indicates the first divergence to the left which will be to the up slow line.

Signal 6 indicates the second divergence to the left which is to the up loop line.

Signal 5 indicates a main route which is the up slow.

Signal 4 indicates the divergence to the left to the up loop line.

Signals to the left of the tallest (main) signal indicate divergences to the left.

Signals to the right of the tallest (main) signal indicate divergences to the right.

This is explained in Section C of the Rule Book.

Use of Multiple Lamp Route Indicators in low speed situations.

Note:

Where speeds are low, such as in station areas, the multiple semaphore junction signal can sometimes be replaced by a single semaphore signal accompanied by a route indicator of the type described later in Section 5.3.

3.3 Subsidiary Signal

This is illustrated in figure 6.

Semaphore Subsidiary Signal Arm

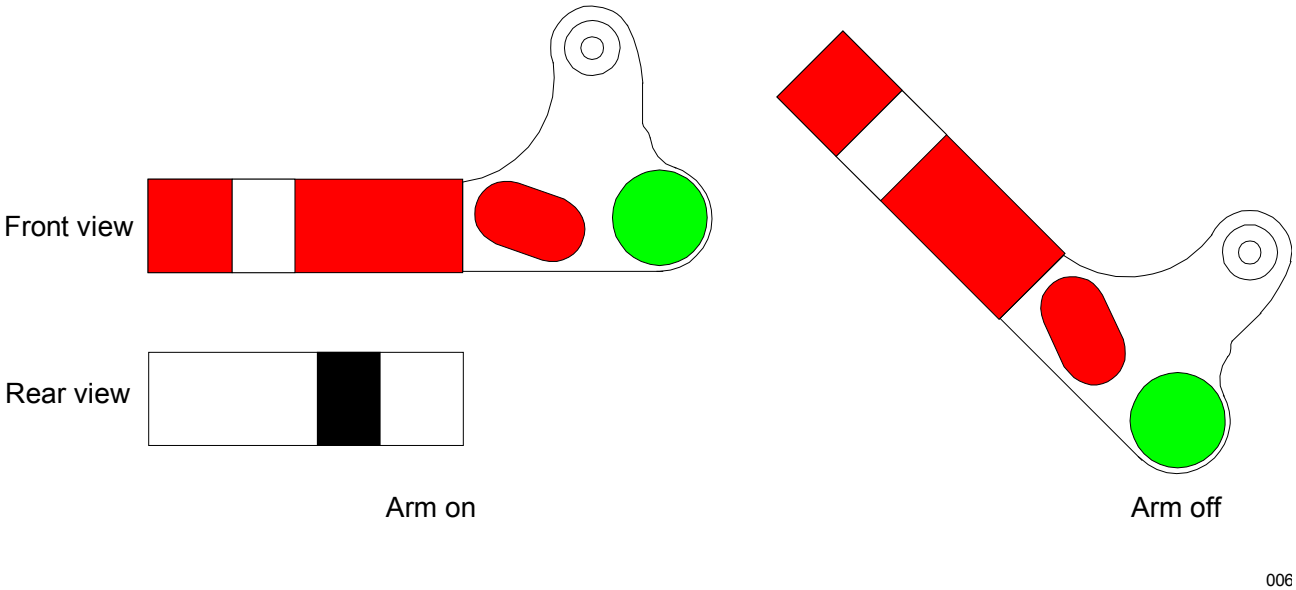


Figure 6

The subsidiary signal consists of a miniature semaphore arm positioned below the main signal arm.

It can be either a

- miniature version of the main stop signal arm as shown in figure 6
- miniature white semaphore arm with horizontal red stripes as shown in figure 7

The subsidiary signal must be read in conjunction with the main signal arm.

When it is in the horizontal position it indicates that the driver should obey the main signal arm.

When the main signal is on and the subsidiary signal is in the off position its meaning is indicated by revealing of the letter C or S.

The letter is hidden when the arm is horizontal and revealed when the arm goes to its off position.

C indicates it is a **calling on signal**.

S indicates it is a **shunting signal**.

Alternative Version of Semaphore Subsidiary Signal Arm

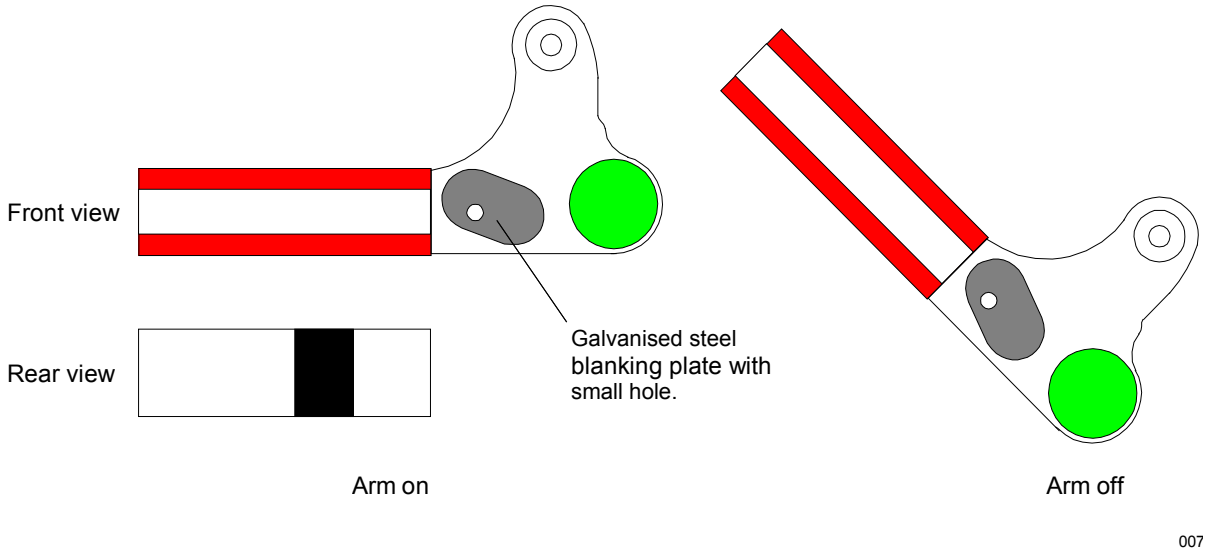


Figure 7

Detailed information concerning subsidiary signals can be found in Section C of the Rule Book.

3.4 Shunting Signals

Shunting signals fall into two categories

- miniature signal arm
- small disc

The miniature arm can be placed

- under a main arm or arms
- on a tall post

The small disc is ground mounted and can be either

- red faced
- yellow faced

Both types of shunting signal can be placed or stacked one above the other. This can be seen in figure 8.

Figure 9 shows how this stacking will indicate specific routes. The top disc or arm indicates the furthest route to the left. The one below indicates the next route from the furthest left and so on.

Signal 1 reads to the up fast line

Signal 2 reads to the up slow line

Each of these will now be examined in greater detail.

Red Faced Disc Signals

Red Faced Discs

Figure 8 illustrates the on and off positions for this signal.

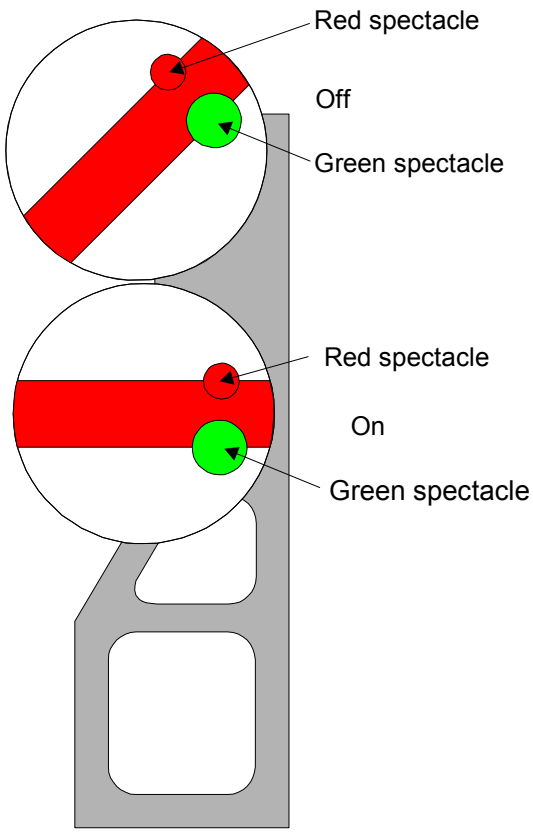


Figure 8

Red Faced Disc Signals indicating specific routes

When the signal is in the **on** position, it is a **stop signal**. When the signal is in the **off** position, it tells drivers that they may proceed as far as the line is clear, or to the next signal.

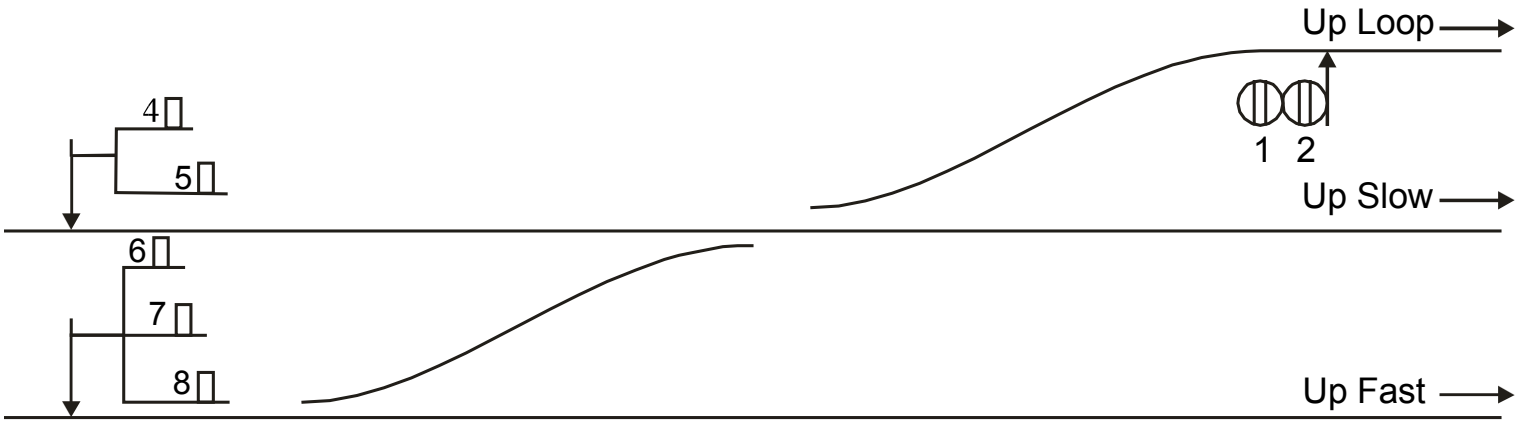
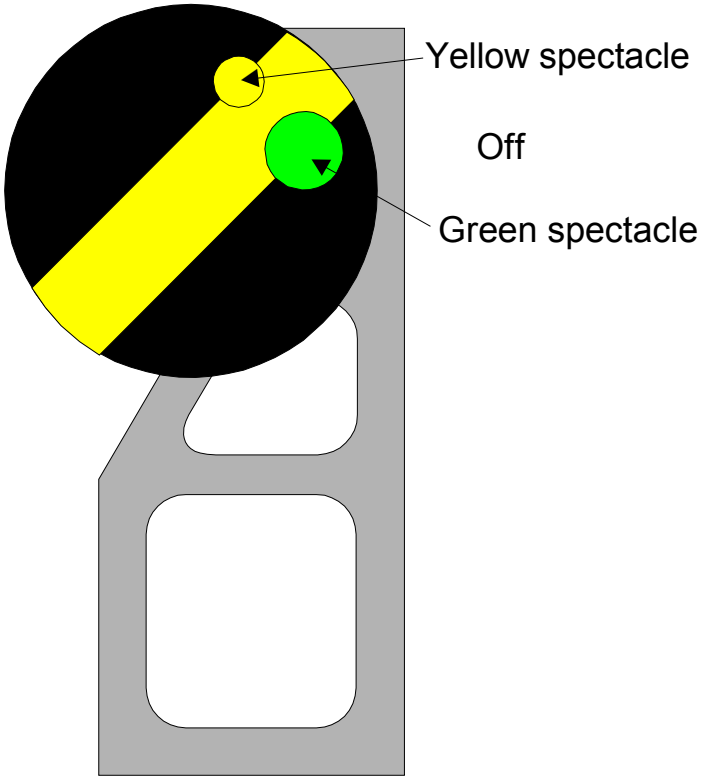


Figure 10

Yellow Faced Discs

Figure 10 provides an example of the yellow faced disc. When the signal is in the **on** position, the driver can proceed for shunting movements. When the signal is in the **off** position, it indicates that the driver is moving out onto a running line.

Yellow Faced Disc Signal

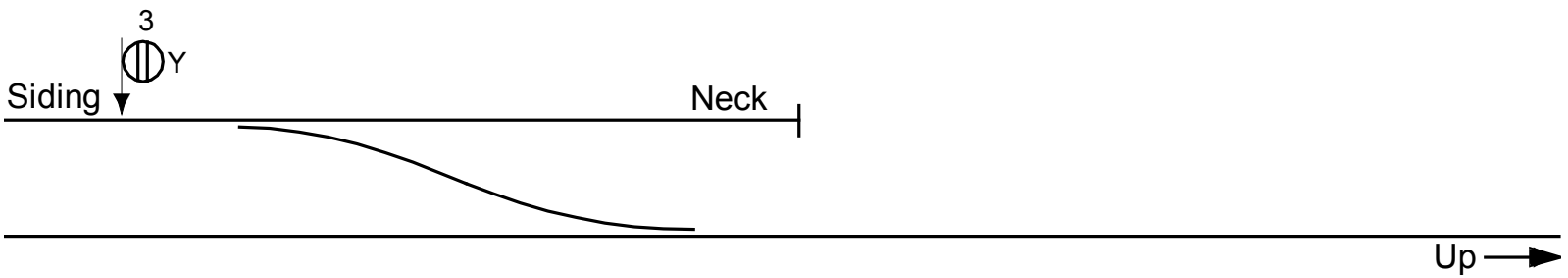


010

Figure 11

Figure 11 illustrates the use of a yellow faced disc.

Utilisation of Yellow Faced Disc Signal



011

When the yellow faced signal is on movements into the Neck line only are allowed. When the signal is off the driver is cleared to move out onto the Up line.

3.5 Section Summary

This section has looked at mechanical signals.

Now complete Self-Assessment Questions 1 to 4 before proceeding to Section Four.

Self-Assessment Question 1

State the two night-time aspects of a mechanical stop signal.

Self-Assessment Question 2

State the information given by an off distant signal to a train driver.

Self-Assessment Question 3

What is the difference between a mechanical shunt signal with red bar and a yellow bar?

Self-Assessment Question 4

A signal post carries a main stop signal and a subsidiary signal below it. The main signal is on and the subsidiary signal is off revealing a letter C. What does this indicate to the train driver?

Now check your answers with the ones given in the Answers Section. Before moving on to Section Four make sure that you fully understand all the information given in Section Three. If there are any areas which you do not understand, re-read the relevant parts of the text.

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4 Colour Light Signals

4.1 Introduction

There are two types of colour light running signals in use

- multi-aspect unit
- searchlight

Each of these will now be examined in greater detail with respect to the component parts of each type and the difference in operation.

4.2 Multi-Aspect Signal

Figure 12 illustrates a typical multi-aspect signal.

Multi-Aspect Signal

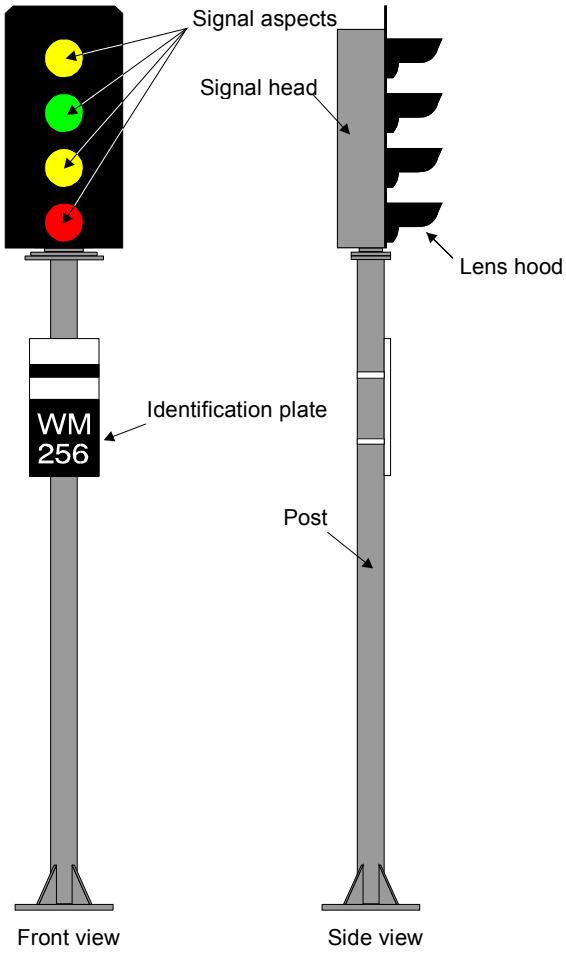


Figure 12

The main components shown in figure 12 include

- signal post
- signal head
- signal aspects
- lens hood
- identification plate

Multi-aspect colour light units may be

- one (single) aspect
- two aspect
- three aspect
- four aspect

Each of these will now be examined in greater detail.

One Aspect Colour Light Unit

The one aspect unit has two practical applications of colour light, namely

- single red aspect
- single yellow aspect

The single red aspect is illustrated in figure 13.

Single Red Aspect

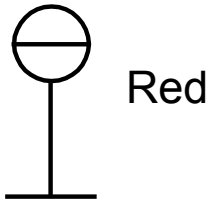


Figure 13

The single red aspect would be used at a station where a platform can be used in **both** directions. The single red aspect indicates the limit of movement in one direction.

The single yellow aspect is illustrated in figure 14.

Single Yellow Aspect

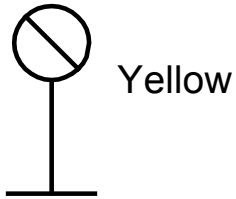


Figure 14

The single yellow aspect is generally used as a fixed distant signal. This may be the distant signal leading into a terminus.

These figures are shown in the form which they would take in a **signalling plan**. All of the diagrams relating to the colour light unit are shown in this form.

Two Aspect Colour Light Unit

The two aspect unit has three **practical** combinations of colour light.

Figure 15 represents two examples of a stop signal.

Two Aspect Stop Signal

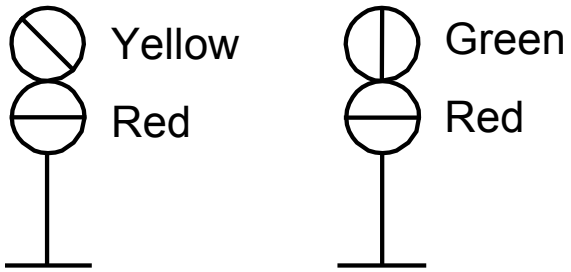


Figure 15

Figure 16 represents a distant signal

Two Aspect Distant Signal

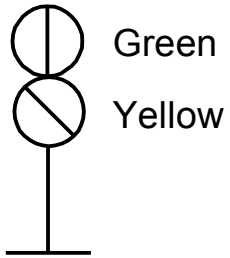


Figure 16

Three Aspect Colour Light Unit

When the line capacity of a two aspect system becomes inadequate due to either

- an increase in the number of trains
- the speed of the trains

then the three aspect system would be used. The three aspect unit can indicate

- red - stop (danger)
- single yellow - clear to next signal (caution)
- green - line clear

If the signal indicates green - line clear, the driver would either

- proceed at full line speed
- proceed at the maximum speed for the train

whichever is the lower.

This can also mean that the next signal is displaying a single flashing yellow.

Refer to Section C of the Rule Book for additional information concerning the single flashing yellow.

Figure 17 illustrates a three aspect stop signal.

Three Aspect Stop Signal

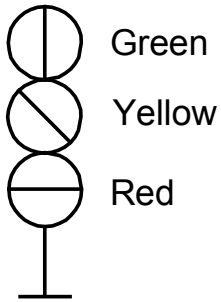


Figure 17

Four Aspect Colour Light Unit

The four aspect system can accommodate more trains than a three aspect system. The efficiency of trains running on mixed traffic lines is greatly improved with four aspect signalling.

Four Aspect Signal

Aspects used on a four aspect unit indicate

- red - stop (danger)
- single yellow - clear to next signal (caution)
- double yellow - next signal at yellow (preliminary caution)
- green - line clear

If the signal indicates green - line clear, the driver would either

- proceed at full line speed
- proceed at the maximum speed of the train
- whichever is the lower.

After a green signal the next signal can be showing green or double yellow or flashing double yellow.

A four aspect unit is shown in figure 18.

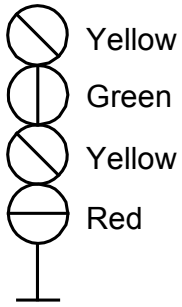


Figure 18

If the signal indicates a double flashing yellow, the driver would either

- proceed at full line speed
- proceed at the maximum speed of the train

whichever is the lower

Refer to Section C of the Rule Book concerning double flashing yellow and single flashing yellow.

Flashing Green

The other aspect that can be displayed is a **Flashing Green**, used where the line speed exceeds 125 m.p.h.

This indicates to the driver that the next two signals in advance are showing green or flashing green.

Range of Colour Light Signals

The range of signals can be seen in figure 19.

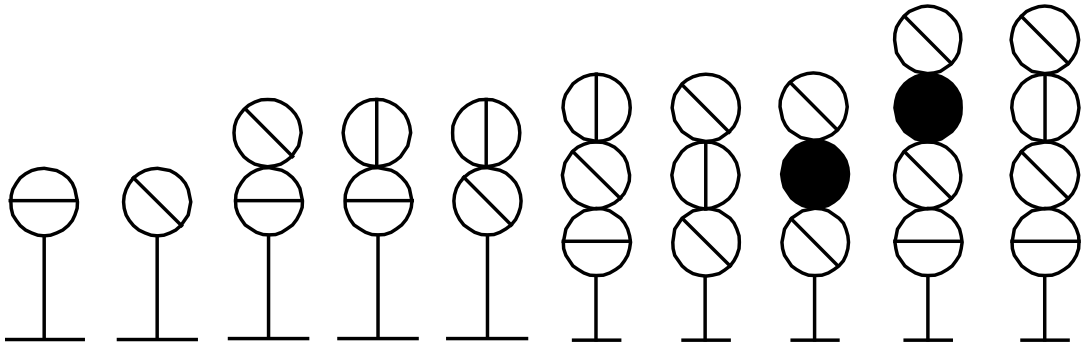


Figure 19

Figure 19 does not include the options that can be created by single and double flashing yellows and by flashing greens.

The aspect that the signal normally displays is indicated by a **double line**. This is illustrated in figure 20.

Four Aspect Signal normally Green

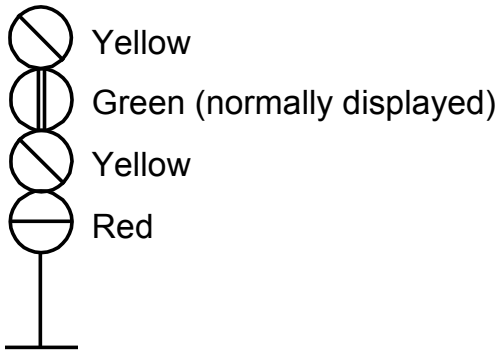


Figure 20

Automatic Signalling

Figure 20 shows a four aspect signal that normally displays green. This arrangement is provided when automatic signalling applies.

Figure 21 shows a three aspect signal that normally displays red.

Three Aspect Signal normally Red

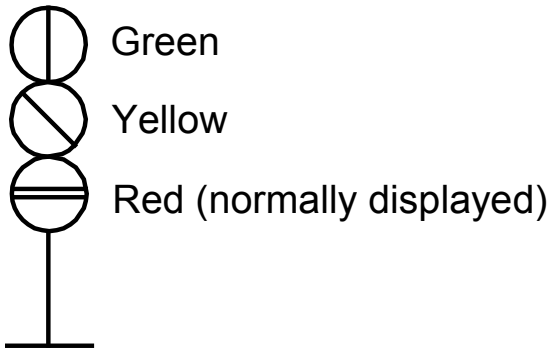


Figure 21

Four Aspect Signal normally Double Yellow

Figure 22 shows a four aspect signal which normally displays a double yellow. This signal is also capable of displaying a **double flashing yellow**.



Figure 22

Figure 22 illustrates the double flashing yellow as indicated by the **four dashes** positioned around the aspect.

At this stage it is worth discussing with your colleagues and mentor the possible locations and uses for each of the signals previously examined. Each signal will be chosen with regard to

- environment
- traffic conditions

Component Parts of a Colour Light Unit

Colour light units are constructed of either

- fabricated aluminium
- cast aluminium

They are sectioned internally for each aspect. Each section is designed to prevent light shining through an unlit aspect. There is an access door at the rear of the unit with the provision for padlocking.

Figure 23 illustrates a side view of a colour light unit which is partially cut away to show the internal components.

Side View of Colour Light Unit (part cut away)

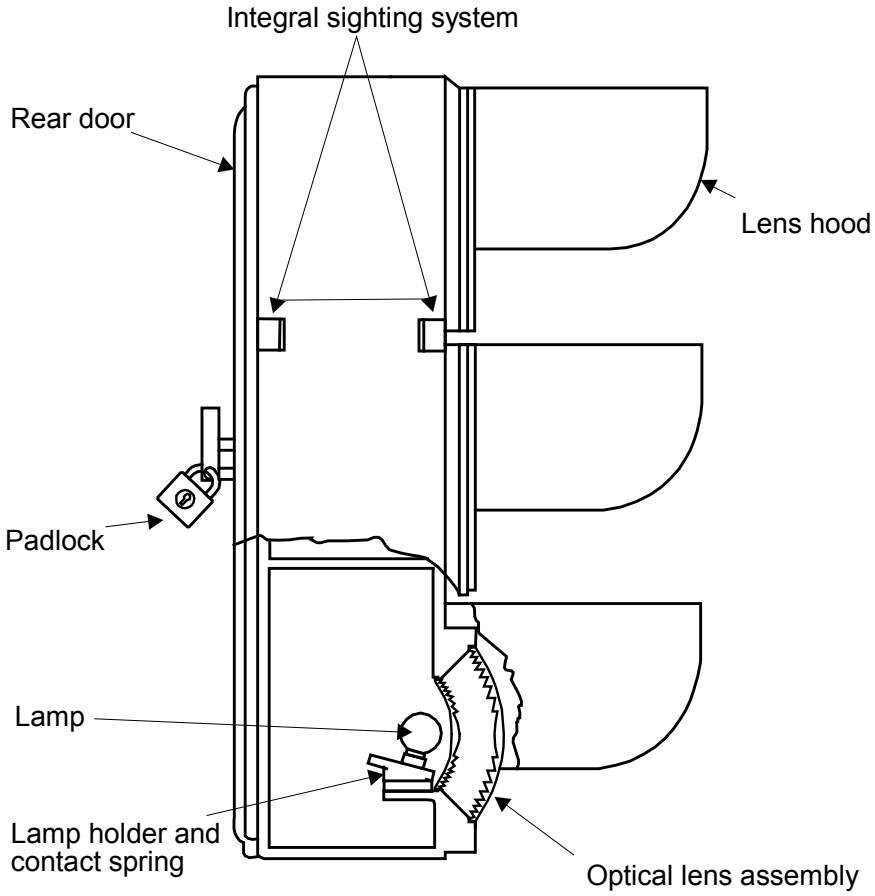


Figure 23

The component parts of the colour light unit include

- lamps
- lamp relays
- lamp holder
- contact springs
- integral sighting system
- optical lens assembly
- transformer

Each of these will be examined in greater detail.

Lamps

There is one lamp for each aspect. In modern signals an SL35 double filament lamp is used. This lamp operates at a nominal 12 volts with each filament being rated at 24 watts. This lamp is identified as

SL35 12V 24/24W.

The two filaments are

- main filament
- auxiliary filament

Construction of SL35 Lamp

Figure 24 illustrates an SL35 lamp showing

- contacts
- connections to each of the filaments

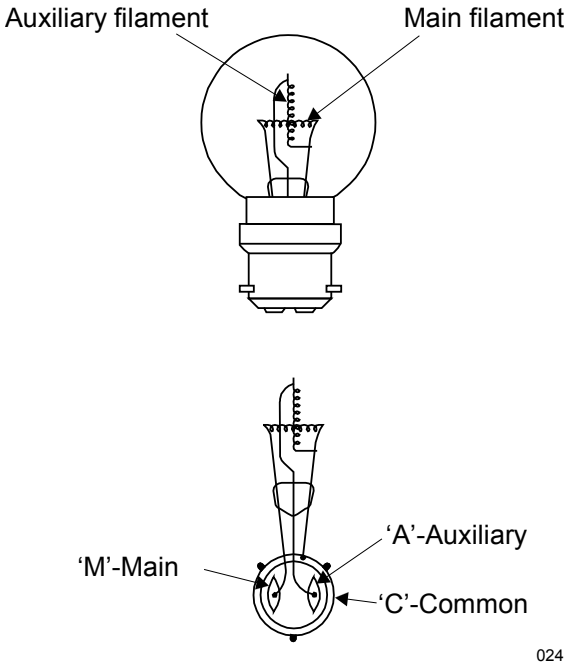


Figure 24

Note that the base of the lamp has **three** pins **unevenly** spaced and therefore will only go in the lamp holder one way.

Lamp Relays

There is one lamp relay per aspect.

The lamp relay performs the following functions

- switching main filament to auxiliary filament in the event of main filament failure
- filament failure indication

Filament proving will be examined in greater detail later in the text.

Lamp Holder

The lamp holder

- is fixed and factory sealed
- holds the lamp in position

The position of the lamp holder is set up in the factory in order to ensure the **filament** is at the **focal point** of the lens system. The lamp holder is then **sealed**.

The seal should not be broken and the lamp holder should **never** be moved.

Contact Springs

There are two springs on the main body of the lamp holder

- one touches the main filament contact
- the other touches the auxiliary filament contact

The common return is made via the lamp cap. This was illustrated in figure 24.

The Integral Sighting System

The integral sighting system allows you to **sight** the signal. This means correctly aligning the beam of the signal.

The Signalling Maintenance Specification states the requirements for the beam alignment of signals. It is worth checking these in order to familiarise yourself with the requirements.

Lens Assembly for Multi-Aspect Signal

Optical Lens Assembly

Figure 25 illustrates the optical lens assembly.

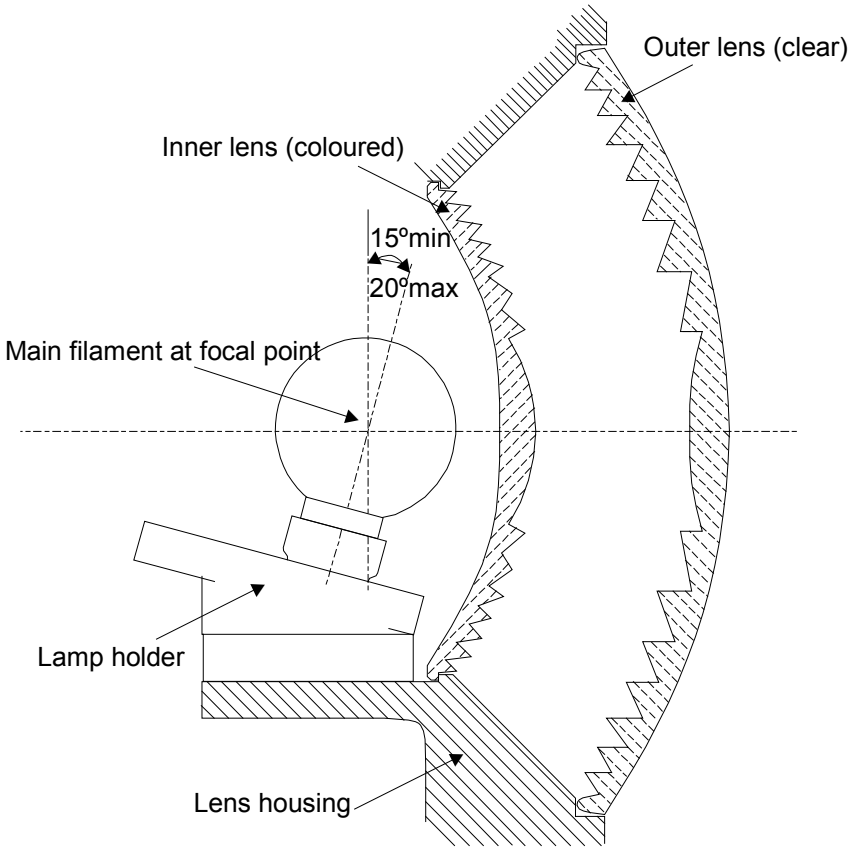


Figure 25

In figure 25 notice that the main filament is situated at the focal point of the lens assembly.

There are two lenses, the

- inner (coloured)
- outer (clear)

The inner lens determines the **colour** displayed.

There are two types of outer lens, the

- long range
- short range

In the case of the long range lens the lens assembly is constructed to allow the light emitted from the clear outer lens to be directed as a narrow beam.

In the case of the short range lens the lens assembly is constructed to allow the light emitted from the clear outer lens to be spread over a larger area. The short range lens is sometimes known as a **spread light lens**.

Hot Strip

Now look carefully at figure 26. This illustrates a **long range lens** which contains a **hot strip**. The hot strip directs a portion of the beam towards the driver when the train is stopped at the signal.

The beam should be aligned as detailed in the Signalling Maintenance Specification.

Illustration of Long Range Lens with Hot Strip

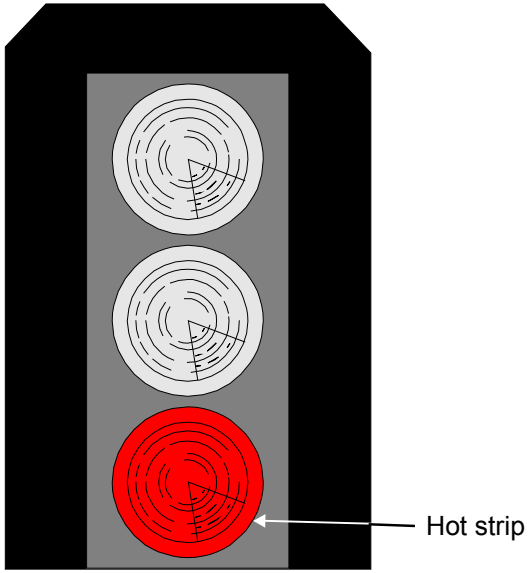


Figure 26

Location of most restrictive aspect

The most restrictive aspect is always placed so that it is nearest the driver's eye level. Normally this means it is at the bottom of the signal head. In this case if there is a build up of snow on the lens hoods the most restrictive aspect is not hidden.

In some circumstances, where there might otherwise be sighting problems, the signal head has to be located below the driver's eye level. In this case the most restrictive aspect is located at the top. Occasionally, where over bridges or station canopies restrict the available headroom, the signal head can be mounted at an angle or horizontally.

The Lens Hood

Lens hoods are provided to prevent **phantom aspects** by shielding sunlight from the aspect. Phantom aspects occur when extraneous light sources cause an aspect to **appear illuminated** to the driver when it is actually extinguished. The Signalling Maintenance Testing Handbook deals with this problem in greater detail.

The Transformer

Each aspect has its own transformer which is used for

- voltage step down
- lamp voltage adjustment

Each of these will now be examined in greater detail.

Voltage Step Down

Each aspect is individually fed by 110 volts a.c. and it is stepped down to a nominal 12 volts by the transformer.

110 volts is used so that signals can be fed over a longer distance with the standard cables. The lamps are rated at a nominal 12 volts therefore the voltages must be stepped down to accommodate this.

Lamp Voltage Adjustment

Lamp Voltage Adjustment

The lamps are actually run at 10.7 +/- 0.2 volts. Adjustment on the transformer allows this to be achieved for both the main and auxiliary filaments.

Operation of the Colour Light Unit

In Module 1, Circuit Reading, the abbreviations used on circuit diagrams were examined. You should be able to

- recognise the specific nomenclature which is used on these diagrams
- locate the various pieces of equipment which are represented

Figure 27 shows the circuit diagram of a four aspect colour light unit. This diagram will be used to explain the operation of the unit.

Circuit for Four Aspect Colour Light Signal

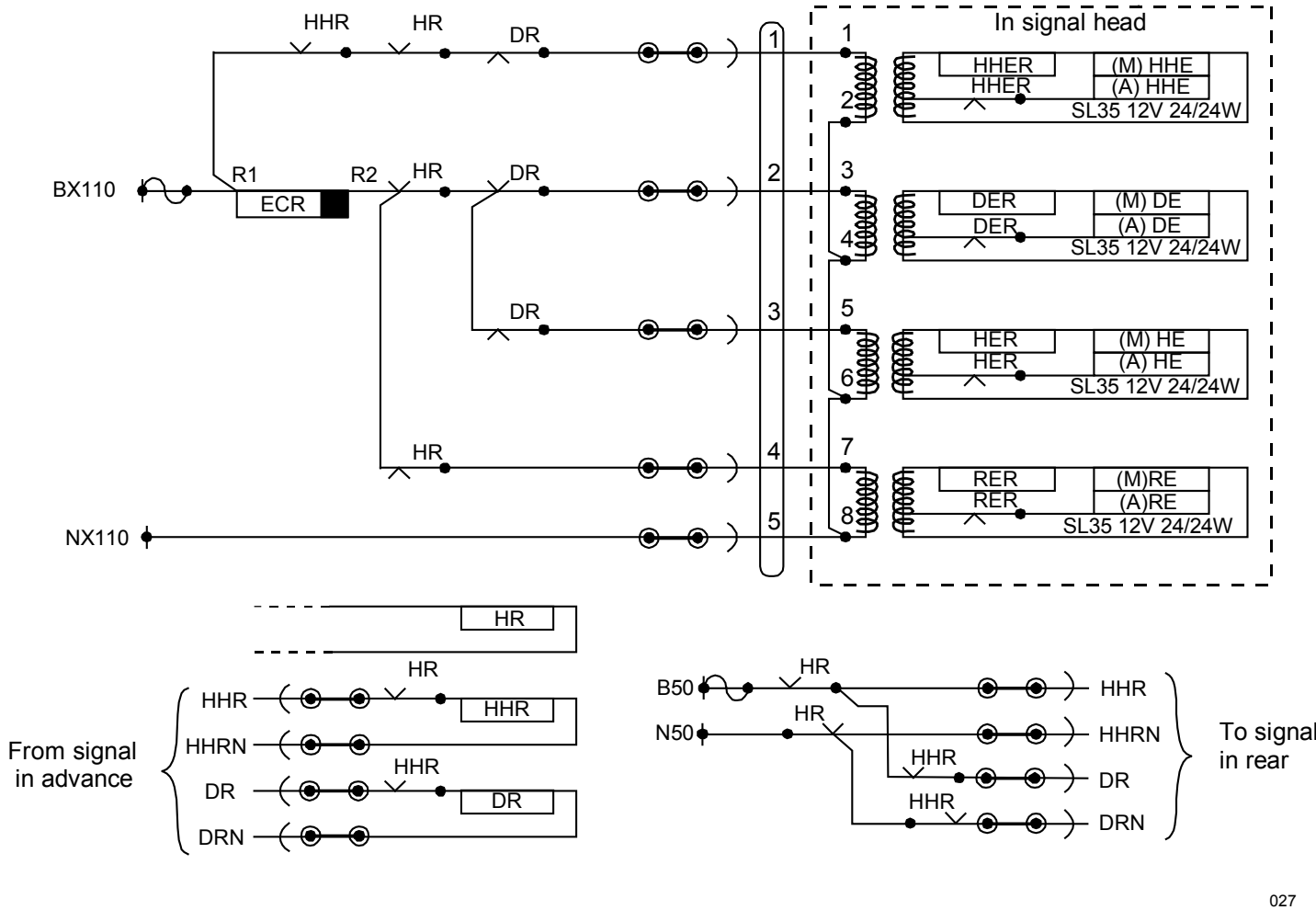


Figure 27

Operation of a Four Aspect Colour Light Unit

Look carefully at figure 27. This circuit illustrates the operation of the four aspect colour light.

Control relays, lamp relays and lamp proving relays

The relays HR, HHR, and DR are known as control relays. The relays HHER, DER, HER and RER are known as lamp relays. The relay ECR is known as the lamp proving relay. An explanation of all their functions now follows.

The control relays and their states when displaying a red aspect in this circuit are

- HR - Yellow relay - de-energised
- HHR - Double yellow relay - de-energised
- DR - Green relay - de-energised

In figure 27 the BX 110 flows via the ECR through the back contact of HR to the signal head transformer for the red aspect and back to the supply by the common NX 110 line.

At the secondary of the transformer the current passes through the RER coil to the main ((M)RE) filament and back to the transformer.

Provided the main filament is intact the RER will **energise**. If the main filament is not intact the RER remains **de-energised** and the current takes the path through the RER back contact and lights the **auxiliary filament ((A)RE**).

With the HR relay de-energised the colour light unit always displays the **red aspect**, ensuring that any fault will produce a **safe** result.

Changing the Aspects

To change the aspect from red to single yellow the **yellow relay** must be **energised.**

- HR - Yellow relay - energised
- HHR - Double yellow relay - de-energised
- DR - Green relay - de-energised

With the HR energised there is no current path to the red aspect because the back contact of HR will open causing the red light to be extinguished and the yellow to become illuminated via the front contact of HR and the back contact of DR.

To change the aspect from single yellow to double yellow, the **yellow relay** and the **double yellow relay** must be **energised**.

HR - Yellow relay - energised

HHR - Double yellow relay - energised

DR - Green relay - de-energised

Note that HHR cannot be energised until HR is energised.

To change the aspect from double yellow to green the **yellow relay, double yellow relay** and the **green relay** must be **energised.**

- HR - Yellow relay - energised
- HHR - Double yellow relay - energised
- DR - Green relay - energised

Note that DR cannot be energised until HR and HHR are energised.

With the DR energised there is no current path to either of the yellow aspects because the back contact of the DR will be open causing both yellow lights to be extinguished and the green to become illuminated via the front contacts of HR and DR.

Filament Proving

Filament proving is achieved via the **front contacts** of the ER (lamp) relays in parallel.

Look carefully at figure 28 which illustrates how filament proving is achieved.

Filaments and Lamp Proving Circuits for Four Aspect Signal

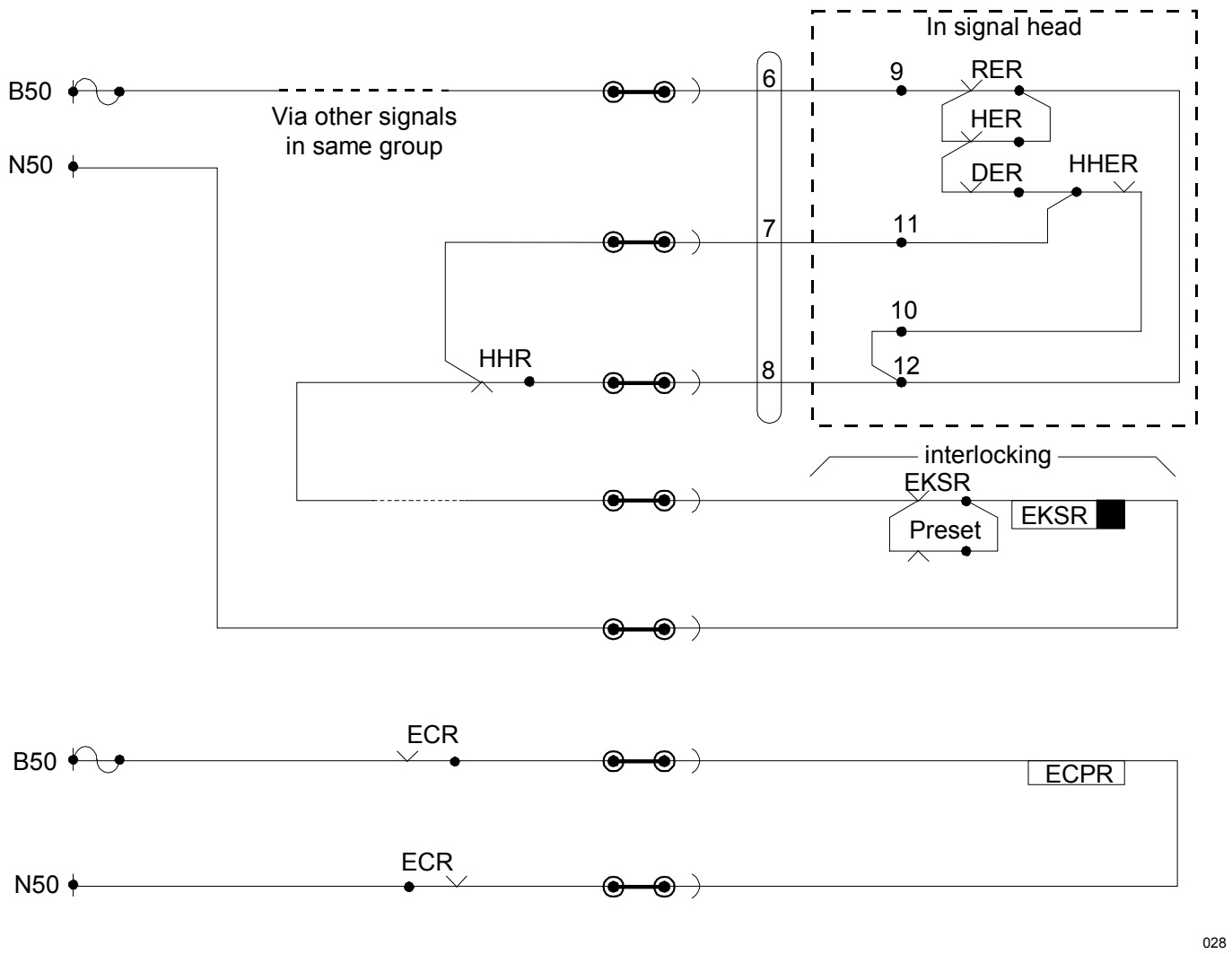


Figure 28

In normal operation a front contact of the ER for the aspect which is **currently being displayed** is **made** providing a path to energise the EKSR at the signal box.

Failure of the main filament would cause the front contact of the ER to be **broken** causing the EKSR to de-energise and provide an indication at the signal box that a main filament has failed within a group of signals.

Lamp Proving

Lamp proving is required because one of the signalling principles states that 'a train should not approach an unlit signal'. Therefore lamp proving is provided in the controls to the signal in the rear so that if the signal is unlit the signal in rear shows red.

The purpose of the ECR (lamp proving) relay previously shown in figure 27 is to prove that an aspect is lit. Failure of both filaments in any one lamp causes the ECR to **de-energise** when that aspect is selected.

In figure 27 the feed to the HHE is not taken via the ECR, which means that should the top yellow fail there would still be a single yellow displayed which is a fail-safe situation.

The ECR relay is a **slow to release relay** for two reasons

- when changing aspects, there is no aspect displayed for a short while
- when a main filament blows and the ER relay is de-energising there will be no aspect displayed for a short while

Under these circumstances the circuitry should not respond to the black signal, as it is only momentarily without a light.

4.3 Searchlight Signals

Figure 29 illustrates typical examples of searchlight signals.

Typical Examples of Searchlight Signals

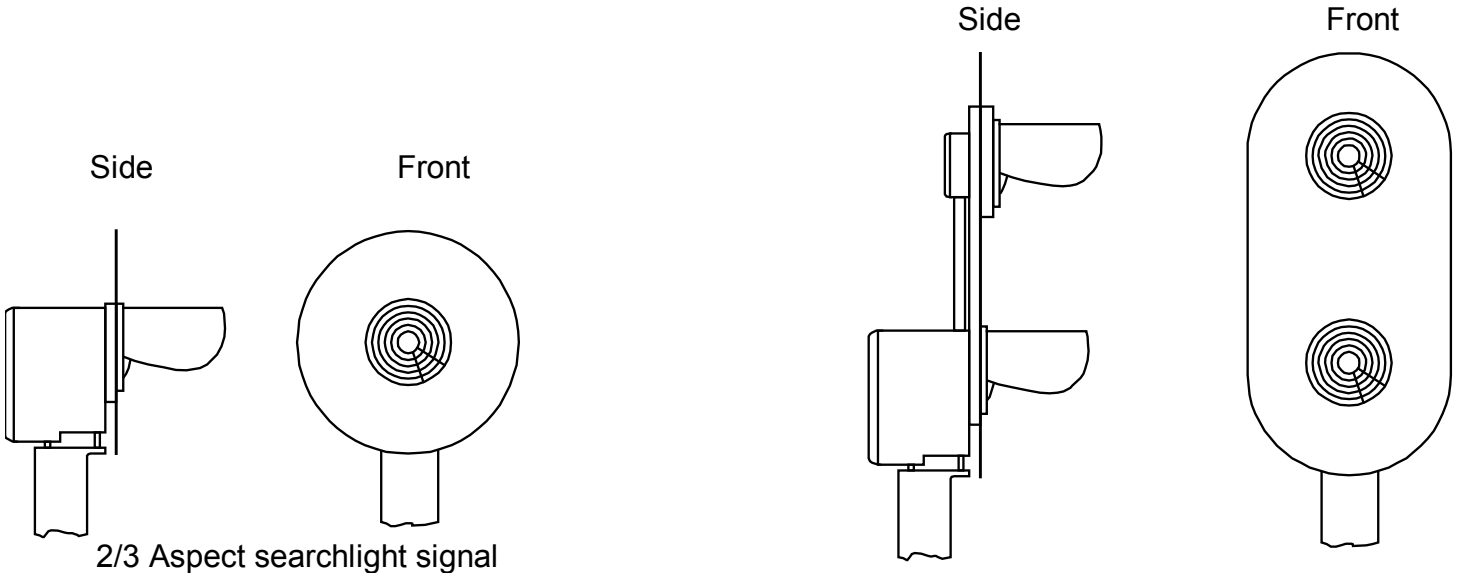


Figure 29

The main component parts of the searchlight signal include

- signal post
- signal head
- lens hood
- identification plates
- signal aspects

Searchlight signal units may be

- two aspect
- three aspect
- four aspect

Each of these will now be examined in greater detail.

Two Aspect Searchlight Signal

The two aspect signal has three practical combinations. Figure 30 represents two examples of a stop signal and figure 31 represents a distant signal.

Searchlight Two Aspect Stop Signals

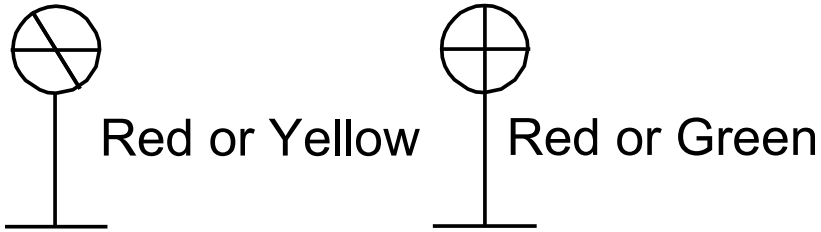


Figure 30

Searchlight Distant Signal

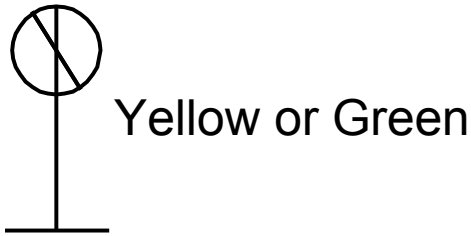


Figure 31

Searchlight Three Aspect Stop Signal

When the line capacity of a two aspect system becomes inadequate due to either

- an increase in the number of trains
- the speed of the trains

then the three aspect system could be used. The three aspect mechanism can indicate

- red stop (danger)
- single yellow clear to next signal (caution)
- green line clear

If the signal indicates line clear (green) the driver would either

- proceed at full line speed
- proceed at the minimum speed for the train

whichever is the lower.

Figure 32 illustrates a three aspect stop signal.

Searchlight Three Aspect Stop Signal

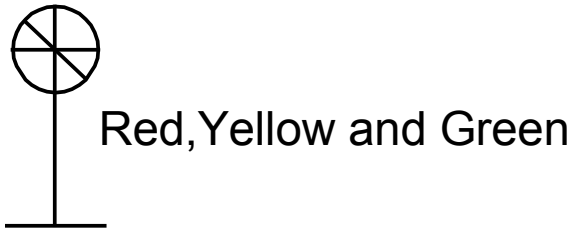


Figure 32

Searchlight Four Aspect Signal

The four aspect system can accommodate more trains than a three aspect signal. The efficiency of trains running on mixed traffic lines is greatly improved with four aspect signalling.

Aspects used on a four aspect signal indicate

- red stop (danger)
- single yellow clear to next signal (caution)
- double yellow next signal at yellow (preliminary caution)
- green (line clear)

If the signal indicates line clear (green) the driver would either

- proceed at full line speed
- proceed at the maximum speed for the train

whichever is the lower.

A four aspect signal is shown in figure 33.

Searchlight Four Aspect Signal

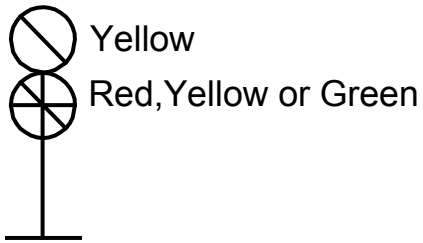


Figure 33

Range of Searchlight Signals

The range of searchlight signals can be seen in figure 34.

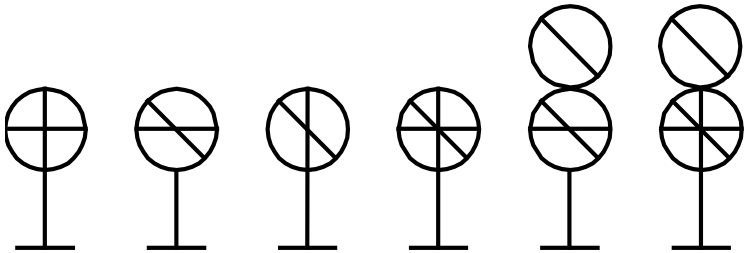


Figure 34

The aspect that the signal normally displays is indicated by a double line. This is illustrated in figure 35.

Normally Green Four Aspect Searchlight Signal

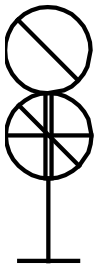


Figure 35

Component Parts of a Searchlight Signal

The searchlight signal consists of the following components

- lamps
- lamp holder/ reflector
- contact springs
- mechanism
- lens assembly

Each of these will now be examined in greater detail.

Lamps

There is one lamp for the mechanism and a second for the top yellow. In searchlight signals an SL32 double filament lamp is used. This lamp operates at 12 volts with the filaments being located one above the other.

- 12 Watts for the main
- 16 Watts for the auxiliary

The lamp is identified as SL 32 12 Volt 12/16 Watt.

Figure 36 illustrates an SL 32 lamp showing

- contacts
- connections to each of the filaments

Construction of SL32 lamp

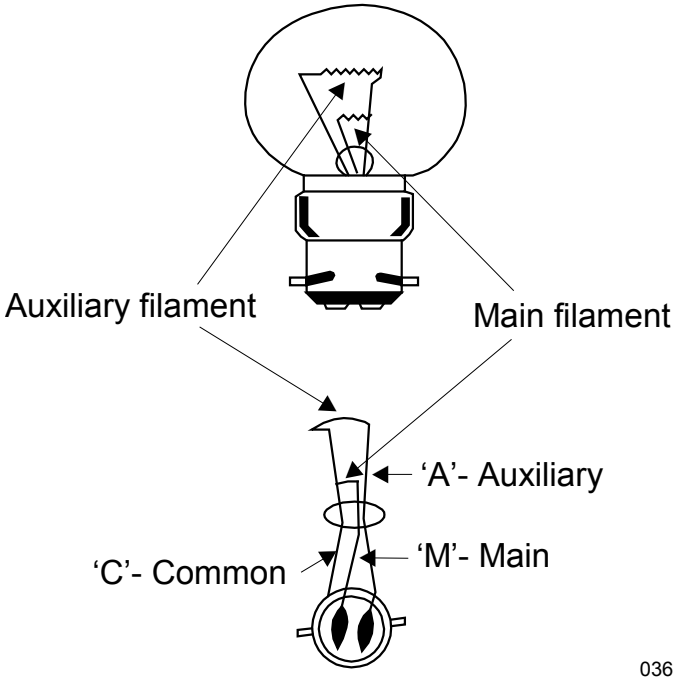


Figure 36

Note that the base of the lamp has **two** pins **evenly** spaced and will therefore go into the lamp holder either normal or reversed. When the lamp is installed it is important to check that the **main** filament is lit.

Lamp Holder/ Reflector

The lamp holder

- is fixed and factory sealed
- holds the lamp in position

The position of the lamp holder is set up in the factory in order to ensure the **filament** is at the inner **focal point** of the reflector. The lamp holder is then sealed. The seal should not be broken and the lamp holder should never be moved.

Contact Springs

There are **two** springs on the main body of the lamp holder

- one touches the main filament contact
- the other touches the auxiliary filament contact

The common return is made via the lamp cap. This was illustrated in figure 36.

Mechanism

Figure 37 illustrates the roundel.

Roundel for Searchlight Signal

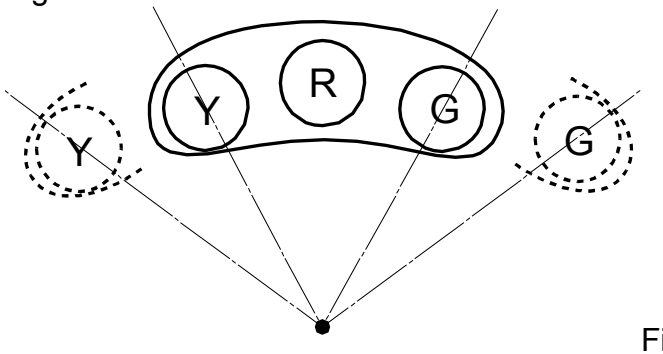


Figure 37

The roundel is housed within the mechanism which is effectively a large relay. The operation of this relay moves the roundel so that the various aspects are displayed. When the relay is de-energised the roundel moves to its middle (red) position by the effect of gravity, that is it fails safe. When the relay is energised with current in one direction the roundel moves to display yellow. Reversing the current moves the roundel to its green position.

Lens Assembly

Figure 38 illustrates the lens assembly of a searchlight signal.

Lens Assembly for Searchlight Signal

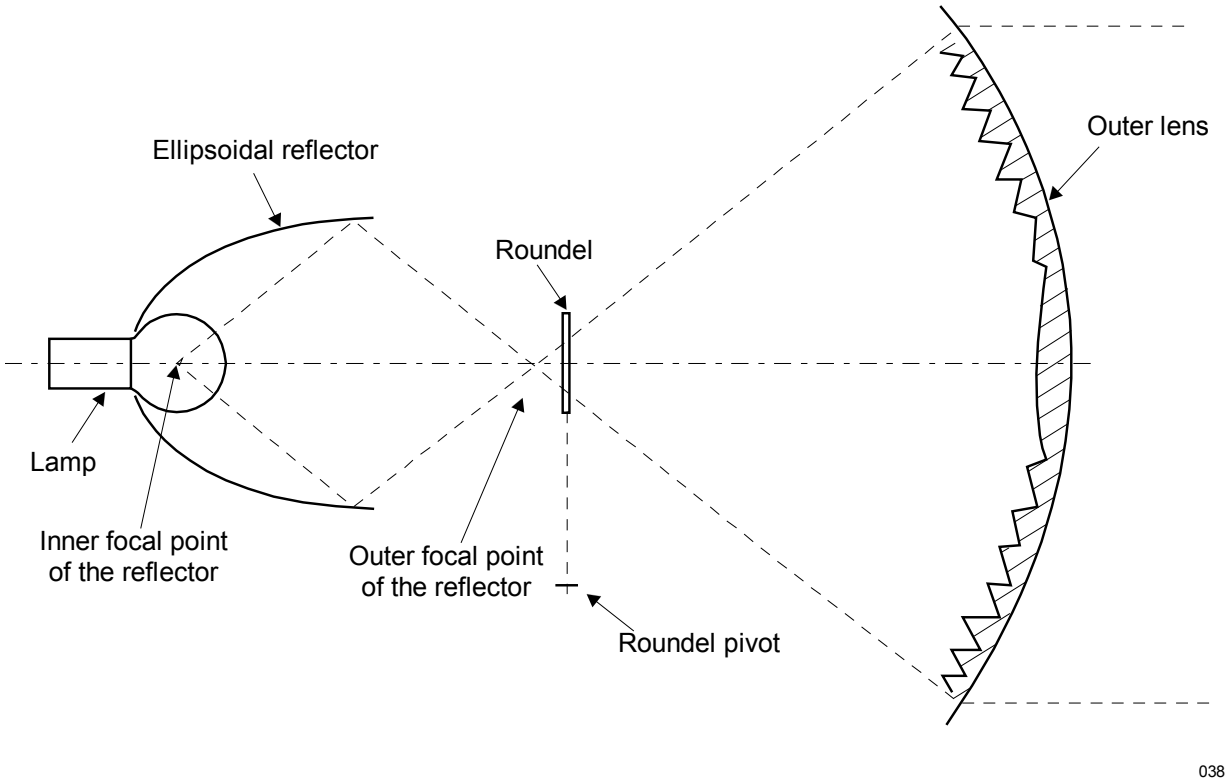


Figure 38

The main filament is focused to shine through the roundel. The outer lens then focuses the emitted light into a narrow beam.

Operation of the Searchlight Signal

Look carefully at figure 39. It shows the control circuit for a three aspect searchlight signal.

Note the following items.

- the portion inside the broken line is within the signal head
- C is the coil of the mechanism relay which moves the roundel
- the contacts RG, HG and DG are moved by the mechanism relay, with the back contacts RG only closed when the roundel is in the red position, the front contacts HG closed when the roundel is in the yellow position and the front contacts DG closed when the roundel is in the green position.
- the relays RGPR, HGPR and DGPR which repeat the status of the roundel in the red, yellow and green positions respectively.

Control Circuit for Searchlight Signal

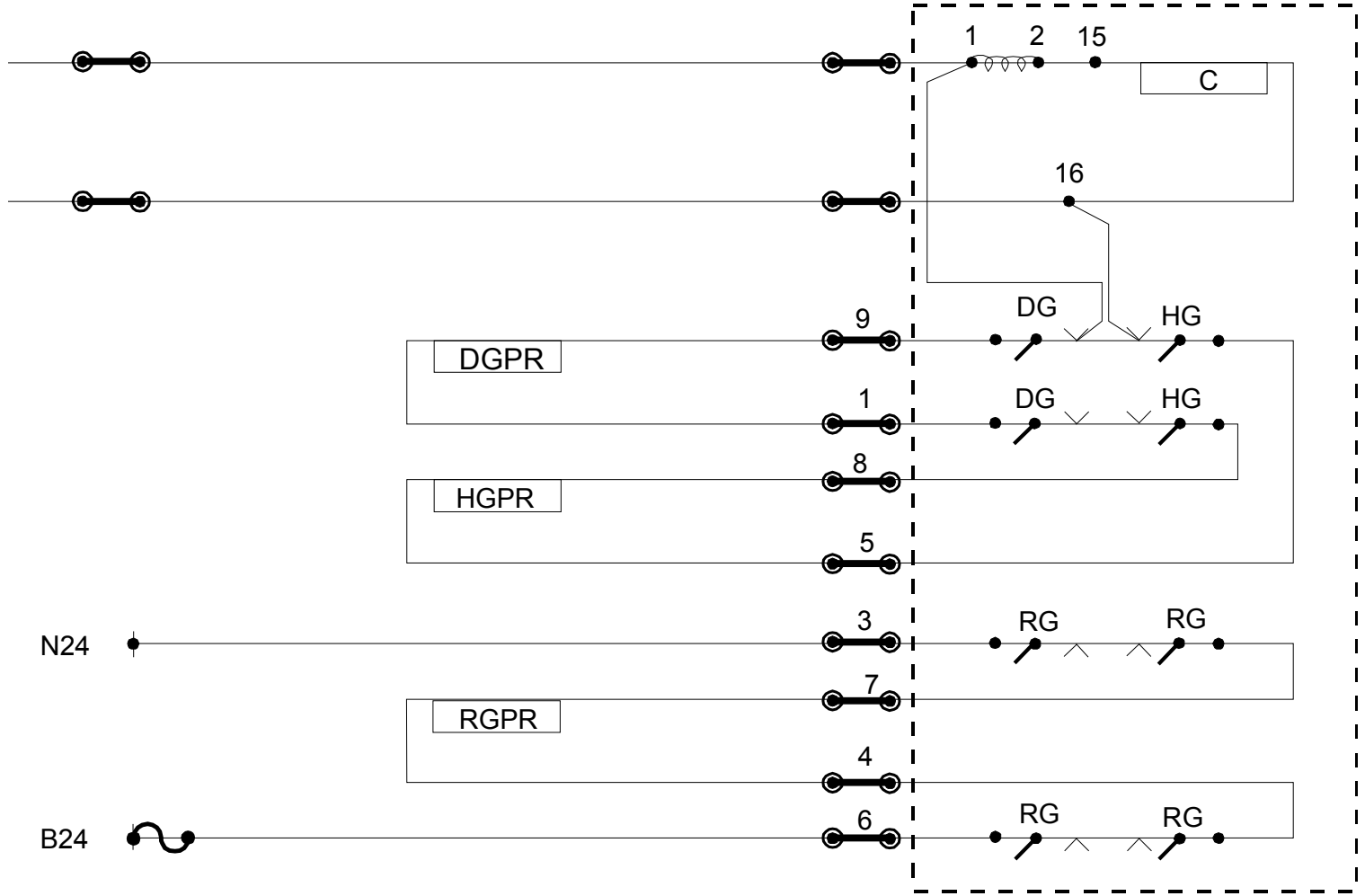


Figure 39

Roundel in Red Position

Now look at figure 40. This shows the energisation of relay RGPR via contacts RG when the mechanism relay C is de-energised. (The conductors connected for B24 are shown in red and those connected for N24 are shown in blue. The de-energised parts of the circuit are shown in black.)

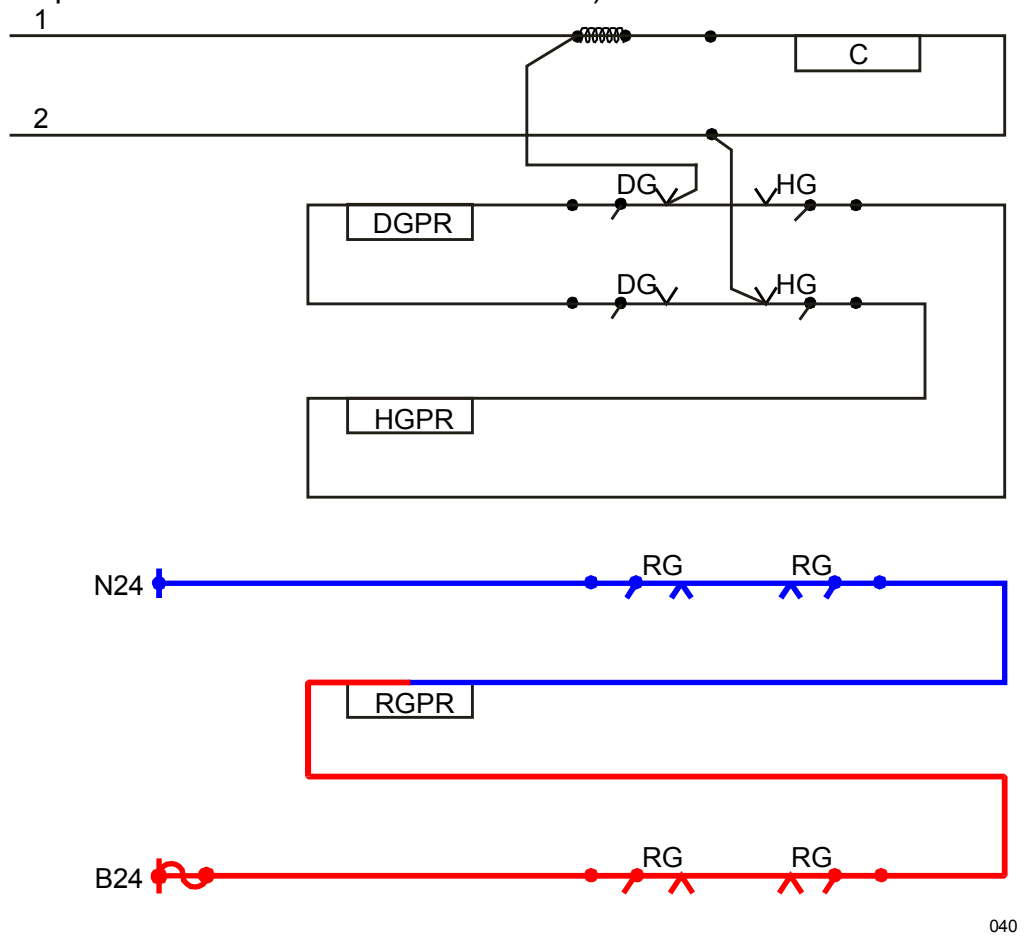


Figure 40

Roundel in Yellow Position

Now look at figure 41. This shows the energisation of relay HGPR via contacts HG when the mechanism relay C is energised with B24 connected to wire 1 and N24 to wire 2.

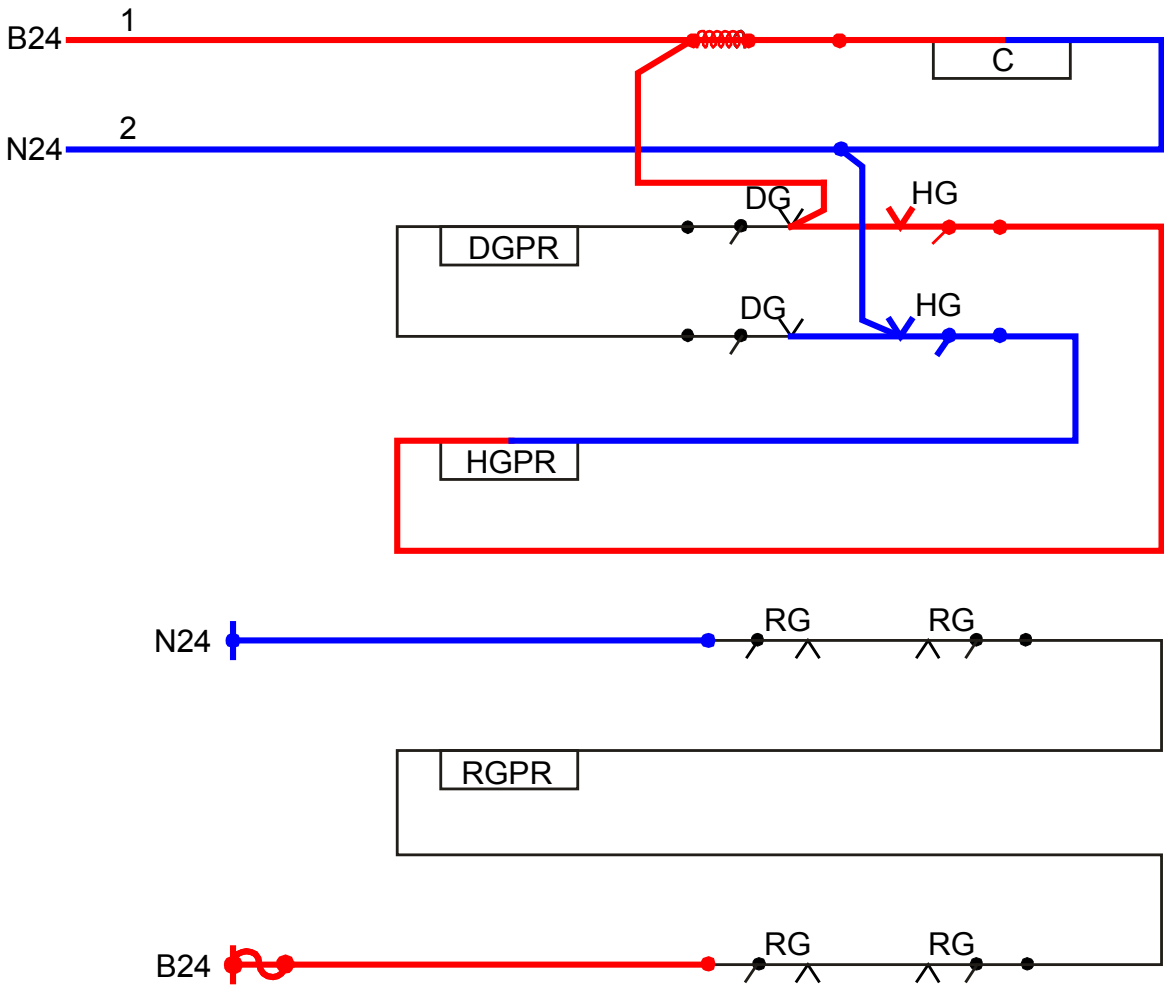


Figure 41

Roundel in Green Position

Now look at figure 42. This shows the energisation of relay DGPR via contacts DG when the mechanism relay C is energised with B24 connected to wire 2 and N24 to wire 1.

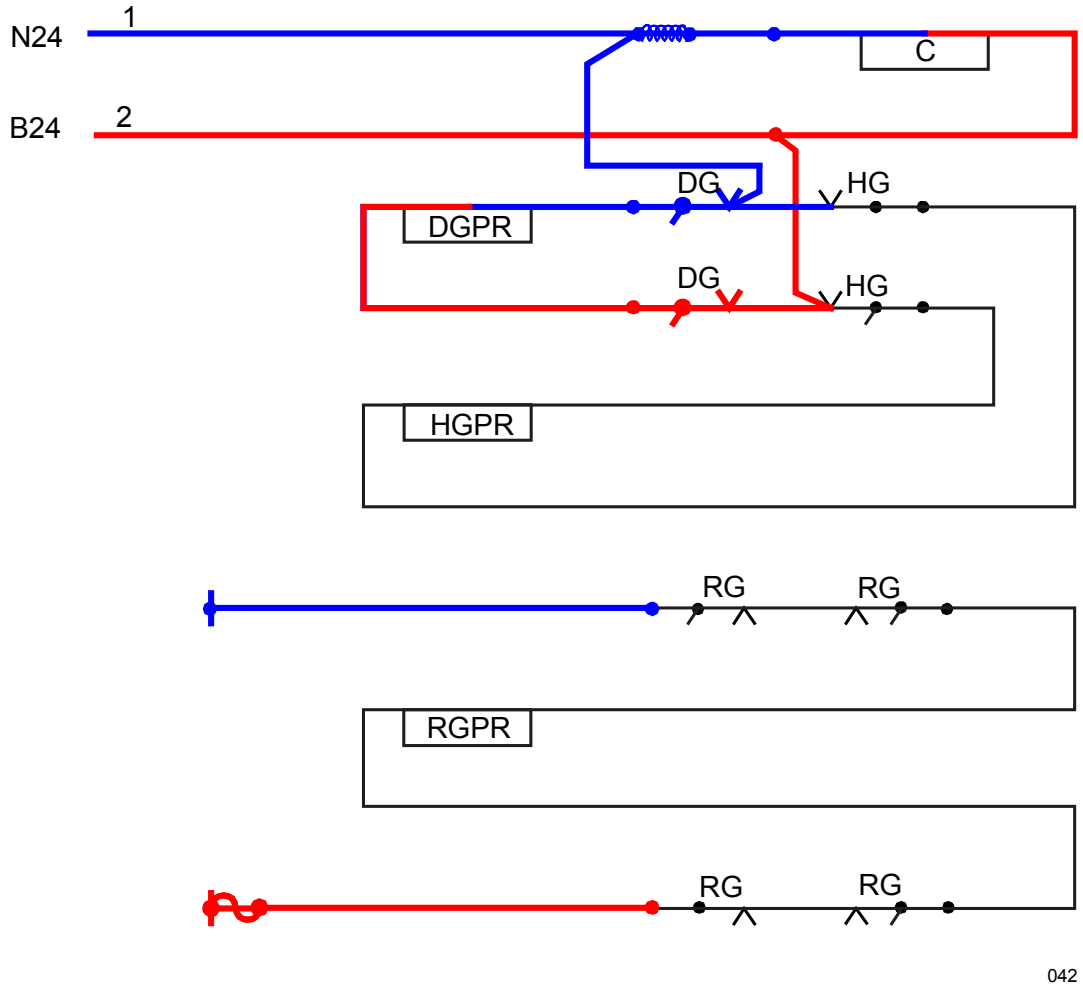
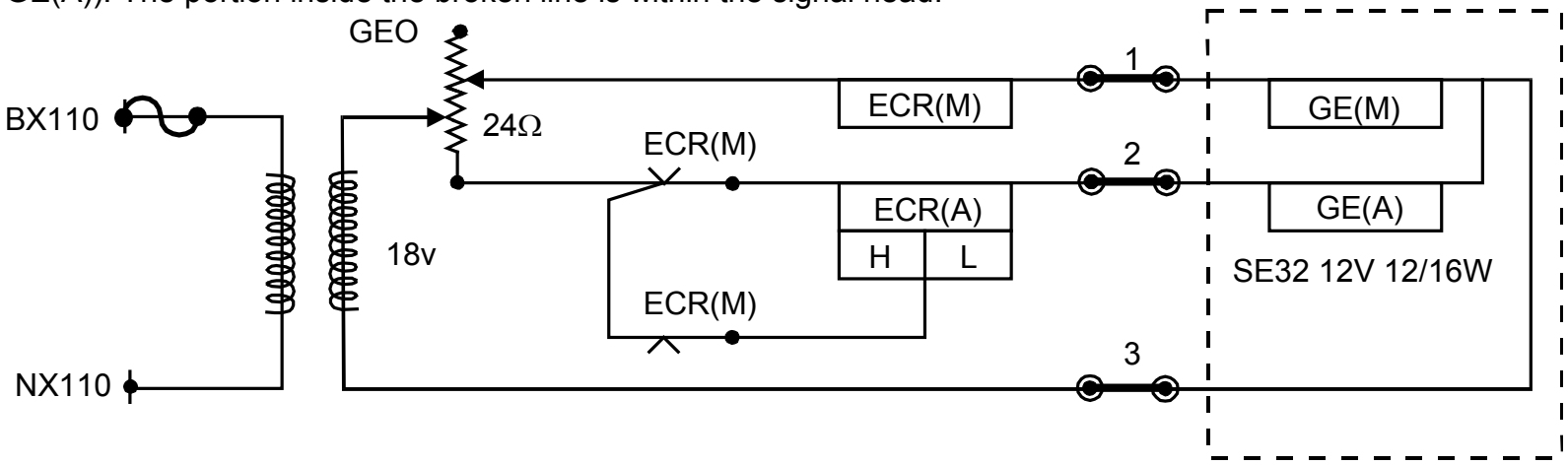


Figure 42

Searchlight Signal Lamp Energisation and Proving

Searchlight Signal Lamp Circuit

Look carefully at figure 43. It shows the circuit which energises the main and auxiliary lamp filaments (GE(M) and GE(A)). The portion inside the broken line is within the signal head.



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Figure 43

The lamp voltage is adjusted (in accordance with the maintenance specification) by adjusting the24 ohm variable resistor GEO. This method compensates for the resistance of the lamp proving relays and the resistance of the tail cable between the location case and the signal head.

Signal Lamp Proving and Changeover

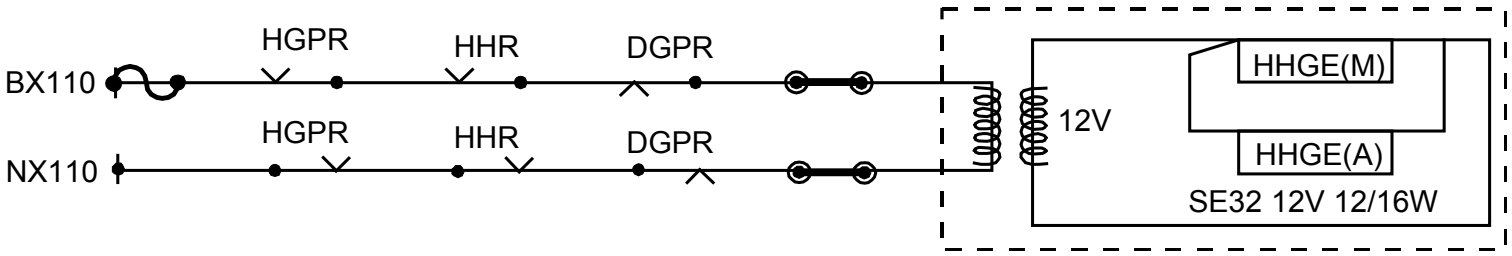
There are two relays dedicated to lamp proving for the searchlight signal. ECR (M) is of normal construction and is energised by the current taken by the main filament. ECR(A) has two windings in series, one high resistance and one low resistance. When the main filament is intact and ECR (M) is energised the ECR (M) front contact is closed and the current to the auxiliary filament has to pass through the high resistance winding. This keeps the current through the auxiliary filament relatively small and does not shorten the life of that filament but does energise the ECR (A). If ECR (A) does not energise it indicates the auxiliary filament is blown.

If the main filament fails ECR (M) drops, its back contact by-passes the high resistance winding and allows the full current to pass through the auxiliary filament. Whilst the auxiliary filament is intact the extra current through the low resistance coil is sufficient to keep ECR (A) energised.

Energising the Top Yellow Lamp

When a four aspect searchlight signal is used, figure 44 shows how the lamp is energised. When a double yellow aspect is required both filaments are lit. No lamp proving is provided, as is the practice with the multiple aspect colour light signals. If both filaments fail before the lamp is replaced during routine maintenance train drivers will report getting a single yellow when they expected a double yellow. This is a fail safe condition. The portion of the circuit inside the broken line is within the signal head.

Circuit for Top Yellow Lamp of a Searchlight Signal



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Figure 44

4.4 Section Summary

This section has looked at

- components of the multi-aspect colour light signal
- operation of the multi-aspect colour light signal
- components of the searchlight signal
- operation of the searchlight signal

Now complete Self-Assessment Questions 5 and 6 before proceeding to Section Five.

Self-Assessment Question 5

Looking at figure 45, explain in your own words how a double yellow aspect is displayed.

Circuit for Four Aspect Colour Light Signal

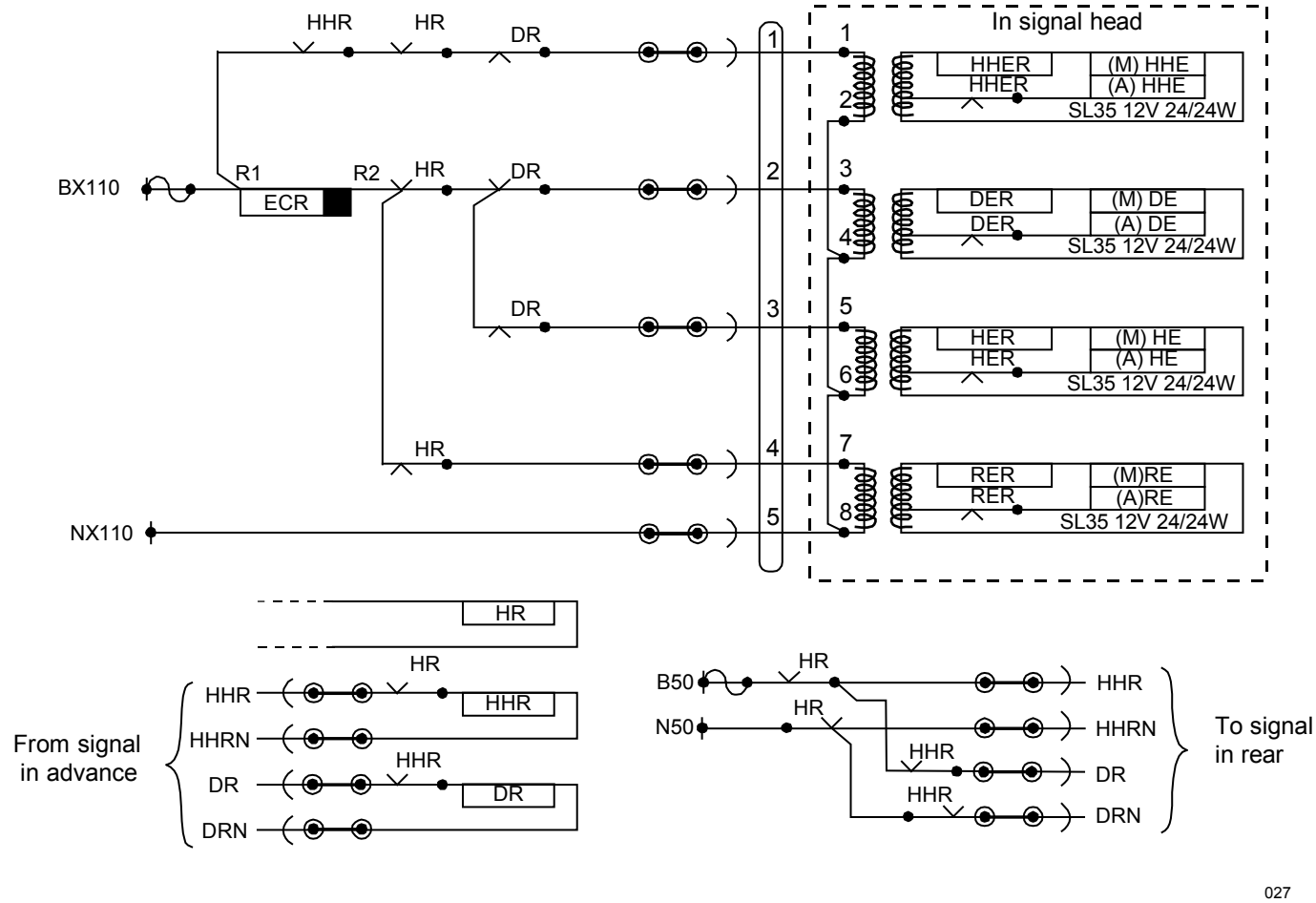


Figure 45

Self Assessment Question 6

Looking at figure 46, explain in your own words the means by which the main and auxiliary lamp voltages are adjusted and how the supply is switched from the main to the auxiliary filament when the main filament blows. Is the coil of relay ECR (A) energised when the main filament is on and how does it detect the condition of the auxiliary filament?

Searchlight Signal Lamp Circuit

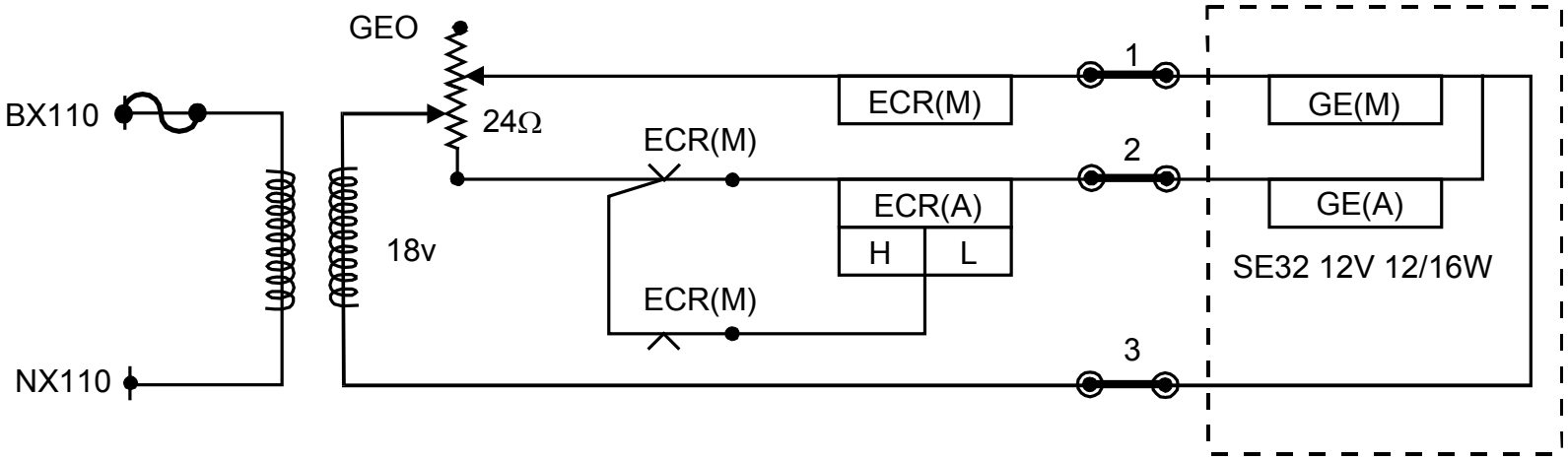


Figure 46

Now check you answers to Self Assessment Question 5 and 6 with the ones given in the Answer Section. Before moving onto Section Five make sure that you fully understand all the information given in Section Four. If there are any areas which you do not understand, re-read the relevant parts of the text.

5 Junction/Route Indicators

5.1 Introduction

Junction/Route indicators

- show the driver the route set
- remove the need for separate signals for each route

There are three main types of junction/route indicators

- Position Light Junction Indicator
- Multi-lamp Route Indicator
- Stencil Type Route Indicator

In this section you will examine the construction and operation of each of these junction/route indicators.

5.2 Position Light Junction Indicator

This is also sometimes called a **feather indicator** and is only used in conjunction with a main colour light signal. It is usually used to indicate when the selected route is a divergence from the **main** or **fastest** route. Normally, no indication is provided for the main/fastest route, but if no obvious main route exists an indication can be given for every route at the junction.

Each indication consists of five white lights in a row. Up to a maximum of six routes can be indicated as shown in figure 47.

Position Light Junction Indicator

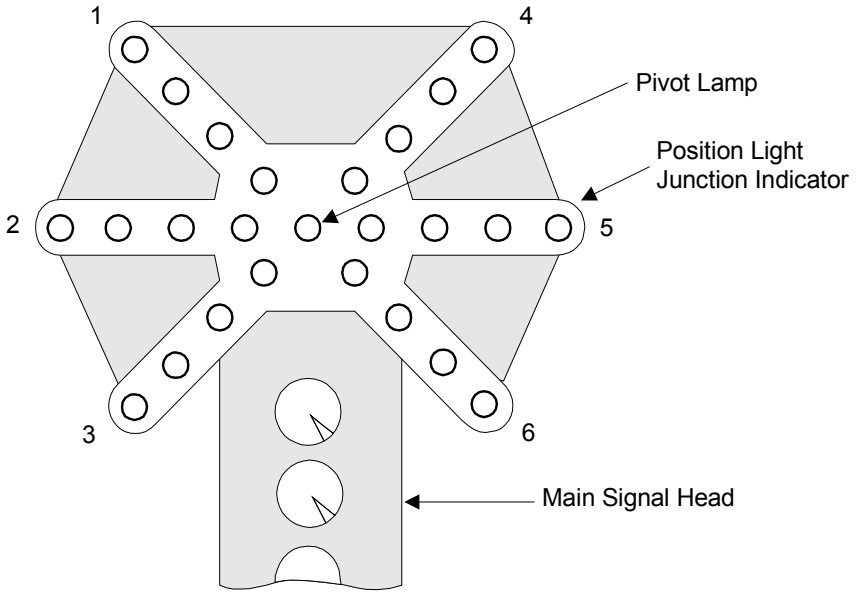
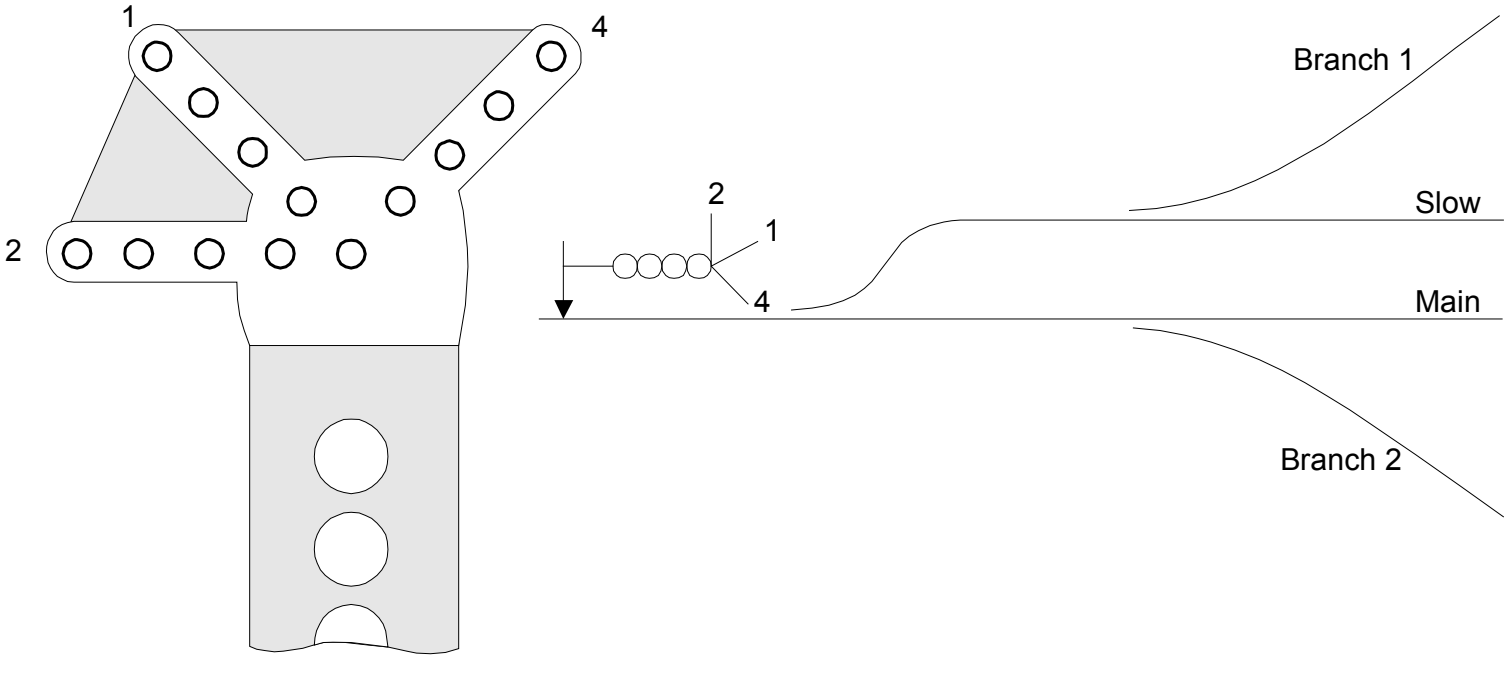


Figure 47

In practice, the **preferred** arrangement of a position light junction indicator has no more than three arms. In this case the arms are never set to form a straight line. Figure 48 gives a typical example and also shows the corresponding signalling plan symbolic form.

Typical Three Route Position Light Junction Indicator



In figure 48

- Position 1 indicates for the route onto the SLOW line
- Position 2 indicates for the route onto the BRANCH 1 line
- Position 4 indicates for the route onto the BRANCH 2 line

When no position light junction indicator is lit this means the MAIN route is indicated.

Line speeds for position light junction indicators

Position light junction indicators are always used to indicate routes where the line speed is in excess of 40 m.p.h. but they can be used for lower speeds as well.

Where the line speeds do not exceed 40 m.p.h. a multi-lamp route indicator (see sub-section 5.3) can be used instead. If the number of left or right turn-outs to be indicated exceeds three, that is beyond the capacity of a position light junction indicator and a multi-lamp route indicator has to be used.

Combined use of indicators

In cases where only some of the routes have speeds in excess of 40 mph, a position light junction indicator can be used in combination with a multi-lamp route indicator.

Component Parts

The component parts of the position light junction indicator will now be examined in closer detail.

Lamps

On older installations the lamps are SL18 12 volt 24 watt single filament. On more recent and new installations the lamps are SL35 12 volt 24/24 watt double filament with only the main filament used.

The lamps are positioned so that the filament is aligned accurately with the focal point of the lens assembly.

Lenses

There are two lenses per lamp

- inner - clear
- outer - lunar white, this filters out yellow light

The lens assembly is used to focus the light into a beam.

Lens Hood

Lens hoods are fitted either to each lamp separately or as a multiple fibre glass unit. The function of lens hoods is to prevent phantom aspects. This has already been explained in Section 4.

Transformer

As with the main colour light signals, the signalling supply to the position light junction indicator is 110 volts AC. One or more transformers are located in the indicator assembly and provide

- voltage step down
- voltage adjustment for the lamps

The lamps are run at less than 12 volts to prolong their life.

Construction of the Position Light Junction Indicator

Where only a single diverging route from the main route exists then the five lamps are mounted in a single casting at an angle of 45° to the left or right of vertical on top of the signal head.

When more than one diverging route is required, a special unit is used to house a common pivot light plus one lamp for each route required. The other three lamps for each diverging route are mounted in additional units which are bolted on as required.

Access to the unit is by means of doors at the rear which should be padlocked.

Operation of the Position Light Junction Indicator

When a single junction indicator unit is in use all five lamps are driven off the same transformer.

In those cases where more than one junction indicator is required the pivot lamp has its own independent transformer and the four other lamps for each route have a separate transformer per route. This is illustrated in figure 49.

Lamp Proving

Figure 49 shows the circuit for two routes of a position light junction indicator.

Circuit for a Position Light Junction Indicator for Routes 1 and 4

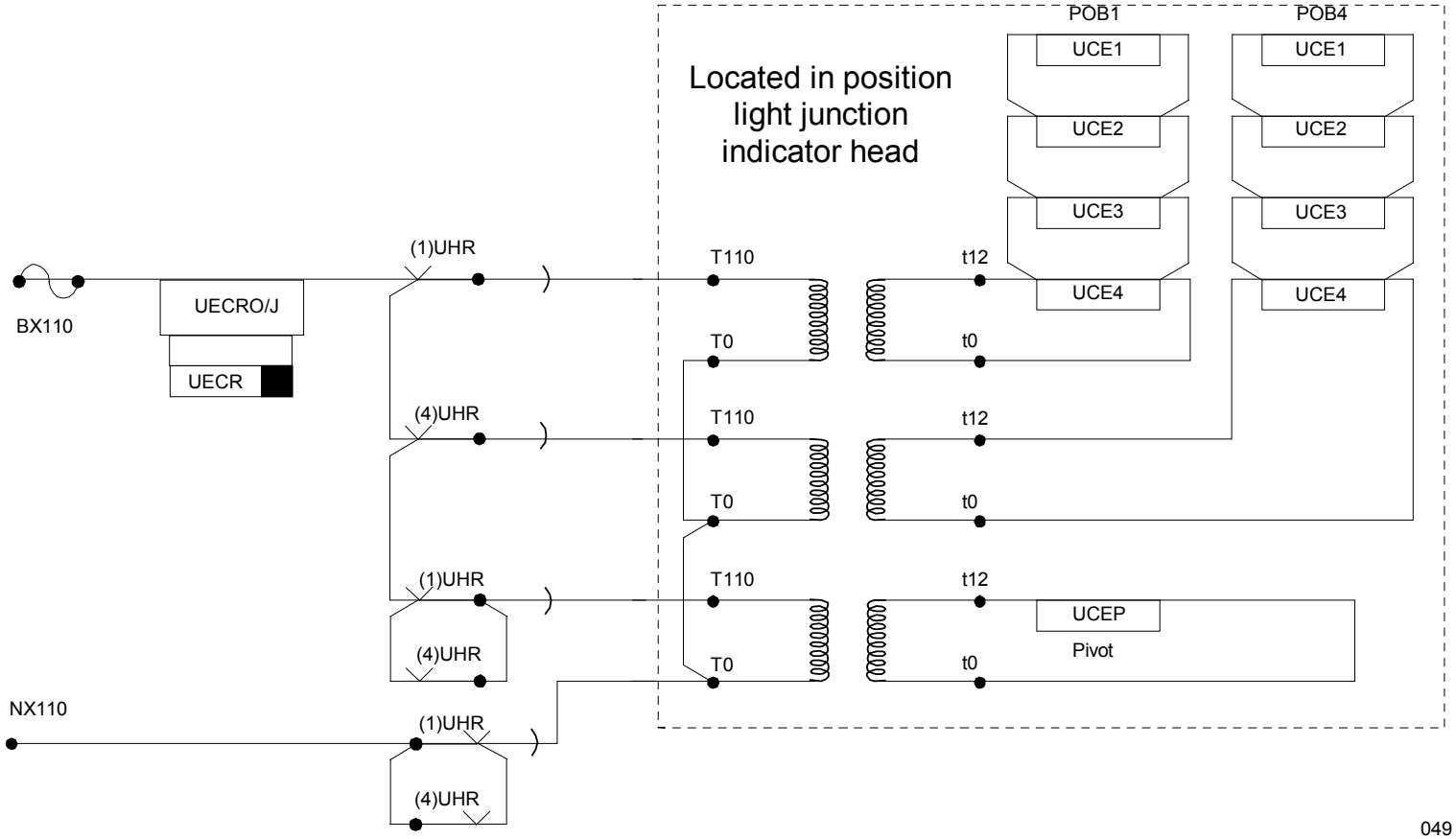


Figure 49

Three lamps must be proved lit

In figure 49 the purpose of UECRO/J is for proving lamp operation.

UECRO/J is a resistor/rectifier unit. The resistance of the unit is adjusted so that the relay UECR will

- energise with three or more lamps alight
- de-energise with two lamps alight

When either of the junction indicators should be illuminated the main signal cannot be cleared until the relay UECR is energised. This prevents any possibility that a driver can approach the signal, find the junction indicator not recognisably illuminated to indicate a lower speed turn-out, assume the train is cleared to proceed on a faster route and consequently run through the turn-out at an excessive speed.

The relays (1) UHR and (4) UHR provide the control to select which junction indicator is illuminated.

5.3 Multi-Lamp Route Indicator

This is also sometimes called a theatre type indicator. It is generally used in colour light signalling areas but is occasionally found with semaphore signals in low speed locations such as stations.

This consists of an indicator showing characters made up from white lights in a 7 x 7 matrix formation to indicate the route to be taken. It can also be a fibre optic unit where fibre optic cables are placed in position within the unit to make up a letter or number.

They are used where position light junction indicators are unsuitable, but can only be used to indicate routes with a maximum speed of 40 m.p.h.

Figure 50 shows the arrangement of the lamps and the case for the unit.

Multi-Lamp Route Indicator

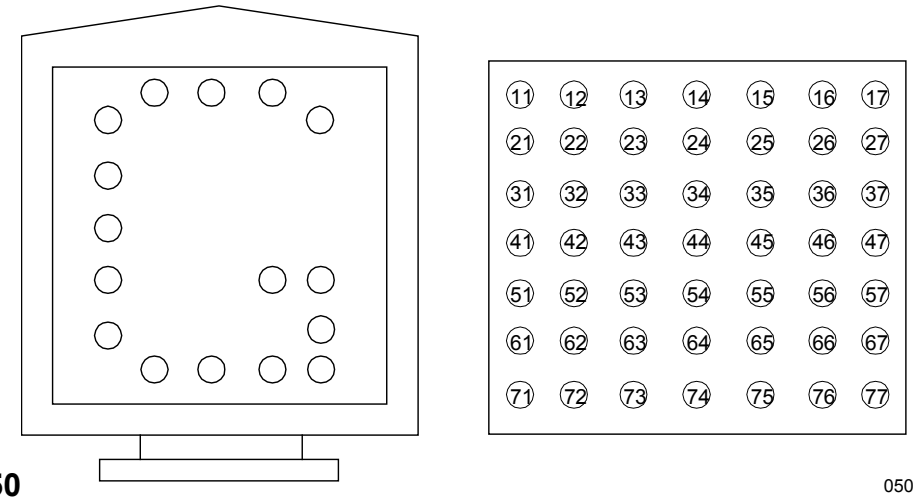


Figure 50 050

Construction of the Multi-Lamp Route Indicator

They are constructed of a metal casting in the form of a box with a glass screen to the front. An access door with padlocking is at the rear. They can be mounted above or to the left of the main aspect signal head.

Hoods are provided for the top and sides to prevent phantom aspects.

The lamps are mounted in bayonet-type holders positioned behind the glass screen and are 110V 15W pigmy lamps.

Operation of the Multi-Lamp Route Indicator

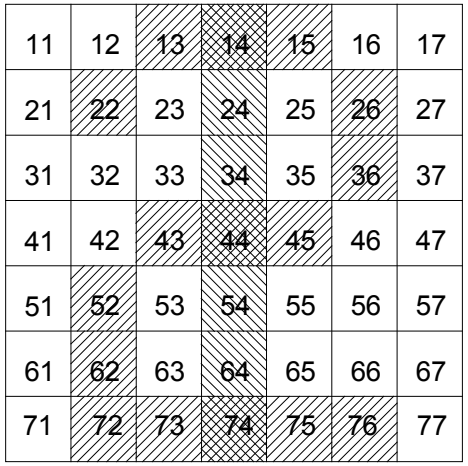
Groups of lamps are wired in parallel and the desired indication is given by selecting the appropriate groups of lamps.

For each possible indication a proportion of the lamps must be proved illuminated before the main aspect can be cleared.

The lamp proving circuit is similar to that for the position light junction indicator but because the indications have different numbers of lamps, adjustable resistors are connected in parallel with the UECRO/J to ensure the same current is taken by UECR for all indications. Sometimes the lamp proving is confined to those lamps which are essential to give a recognisable indication and the condition of the less essential lamps is ignored.

Typical Arrangement of Lamps to show 1 or 2

Figure 51 shows the lamps which must be illuminated to give the indications 1 and 2.



Lamp	1	2
13		X
14	X	X
15		X
22		X
24	X	
26		X
34	X	
36		X
43		X
44	X	X
45		X
52		X
54	X	
62		X
64	X	
72		X
73		X
74	X	X
75		X
76		X

051

Figure 51

Three of the lamps should be lit to either show 1 or 2. Four lamps are lit only to show 1, but thirteen other lamps are required to show 2. The required circuit is shown in figure 52.

Control and Lamp Proving Circuit for Multi-Lamp Route Indicator

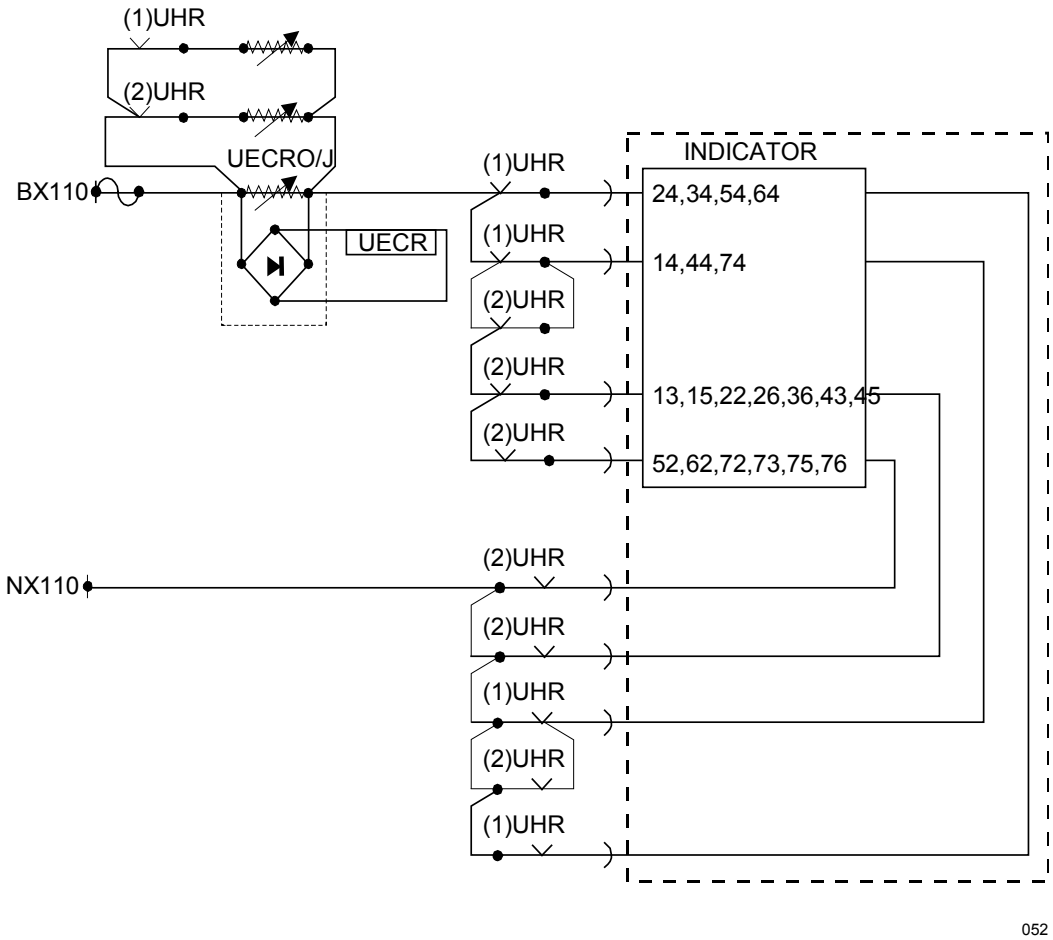


Figure 52

To illuminate the indicator, relay (1) UHR or (2) UHR (not illustrated) must be energised. Workout for yourself how the control and lamp proving is achieved.

5.4 Stencil Type Route Indicator

Stencil type route indicators display a fixed message for the route to be taken. The message can be one or two letters or numbers or a short word. They are used only for low speed movements such as shunting, or in situations where the train is stationary when the driver needs to read the indicator.

Construction of the Stencil Type Route Indicator

They are constructed in the form of a metal box with a glass or perspex screen to the front. Access is at the rear by a door which is held closed by a catch (not locked).

A stencil type route indicator usually uses two lamps (110 V 40 W), wired in parallel which are lit and shine through the stencil plate. If alternative messages are required then additional boxes must be added.

Operation of the Stencil Type Route Indicator

Figure 53 shows a typical circuit for stencil type route indicators.

Typical Circuit for Stencil Route Indicators

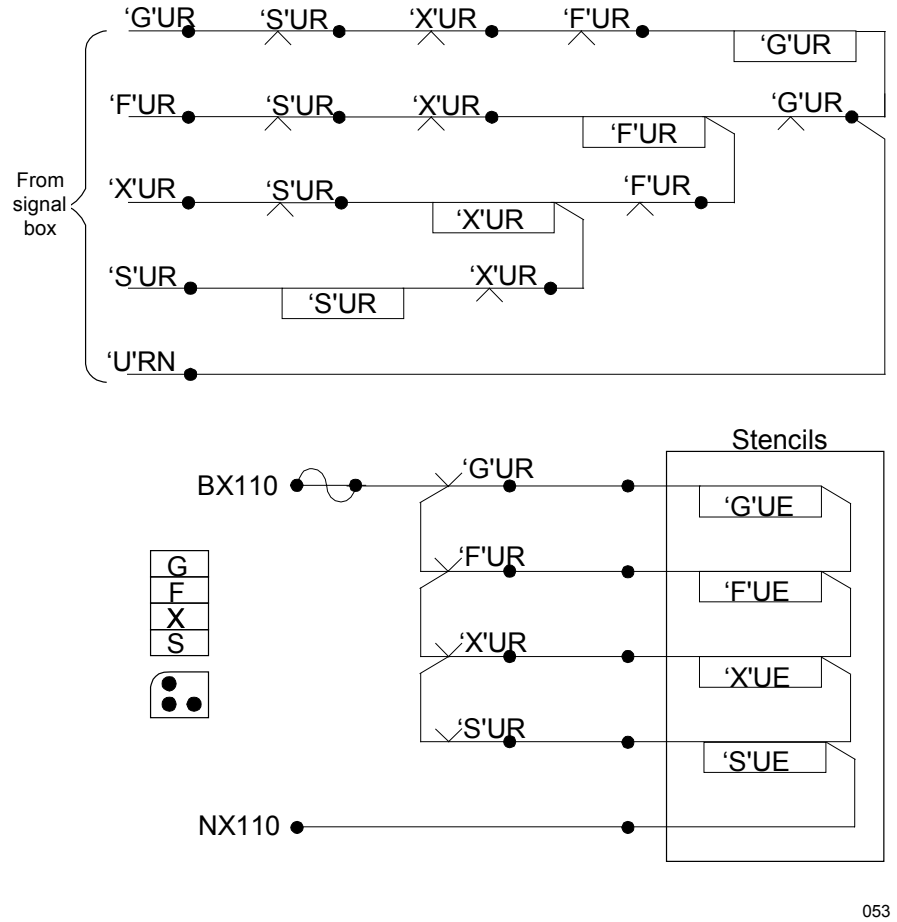


Figure 53

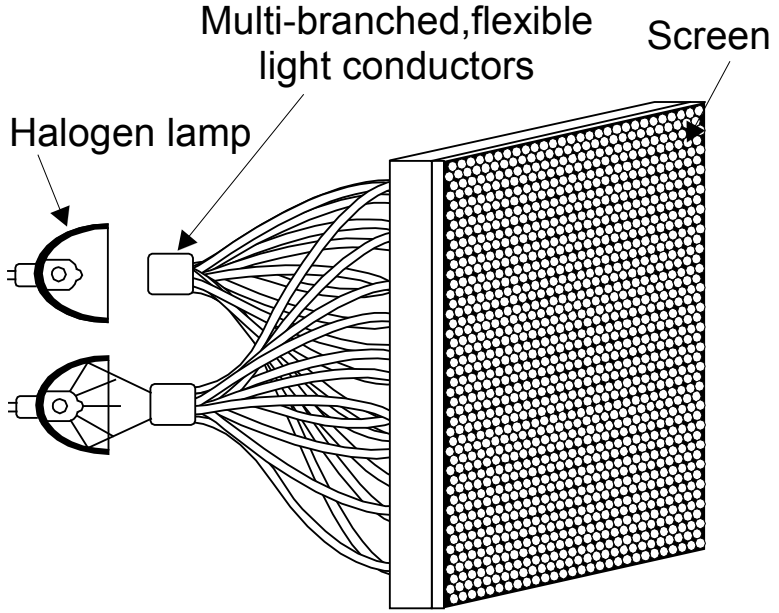
In figure 53 the circuit diagram shows the four route relays (UR's). To energise any one of the UR's the others are proved de-energised, so that the driver of an approaching train is not confused.

Notice that in this case four stencil type route indicators are being controlled. The circuit ensures that only one stencil can be lit at any time. In accordance with the normal practice, lamp proving is not provided.

5.5 Fibre Optics

Theatre and stencil type route indicators can use fibre optics to produce the required display. In this case light from a quartz halogen lamp is focused onto a bundle of fibre optics which produce the required display on the surface of the indicator.

Figure 54 shows the arrangement.



Construction of a Fibre Optic Route Indicator

Figure 54 054

5.6 Section Summary

This section has looked at the

- position light junction indicator
- multi-lamp route indicator
- stencil type route indicator

Now complete Self-Assessment Question 7 before proceeding to Section Six

Self-Assessment Question 7

In a position light junction indicator, how many lamps need to be alight for the main signal to display a proceed aspect?

Now check your answer with the one given in the Answers Section. Before moving on to Section Six make sure that you fully understand all the information given in Section Five. If there are any areas which you do not understand, re-read the relevant parts of the text.

6 Position Light Signals

6.1 Introduction

There are two types of position light signals

- position light subsidiary signal which has two lamps and is associated with a main aspect.
- position light ground signal which has three lamps and is used only for shunting purposes.

Each of these will now be examined in greater detail.

6.2 Position Light Ground Signal

An example of a position light ground signal is shown in figure 55.

Position Light Ground Signal

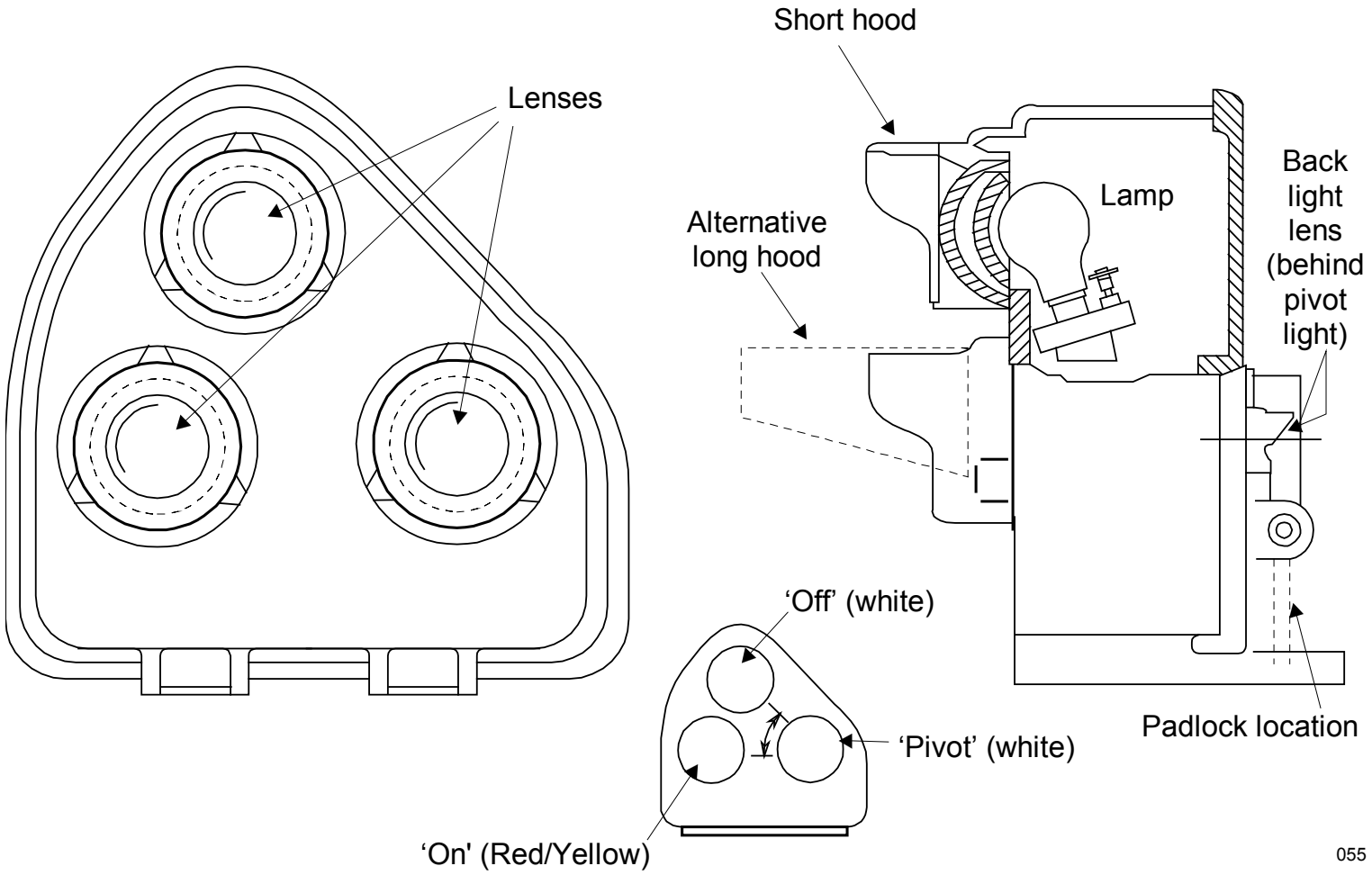


Figure 55

Construction of a Position Light Ground Signal

It consists of three lamps mounted in a metal casting which is sectioned off internally for each lamp. Access is by means of a door at the rear with provision for padlocking.

As the name implies this signal is usually mounted close to the ground on a short concrete post, which has a sloping top to angle the lenses upwards towards the driver's eye. In cases where there would be visibility problems with the signal at ground level, it is mounted on a post at driver's eye level.

Each lamp unit has a double lens assembly. The inner lens is either clear, red or yellow depending on the function of the lamp. The outer lens is lunar white (blue) for the pivot and OFF indication and clear for the ON indication. The function of the lens is to focus the light from the lamp into a beam. Hoods are used to prevent phantom aspects. A back light lens is provided behind the pivot light.

The signal lamps are SE 163 110 volts 35 watts with three pins which fit in the lamp holder so that the filament is aligned with the focal point of the lens assembly.

Operation of Position Light Ground Signal

The indications given by the unit are

ON - One white light known as the pivot illuminated at the lower right hand corner and either a red or yellow light (depending on requirement) in the lower left hand corner.

OFF - The pivot light and a second white light at an angle of 45° above and to the left of the pivot light.

Figure 56 shows the circuit which would be used with a position light ground signal.

Circuit for a Position Light Ground Signal

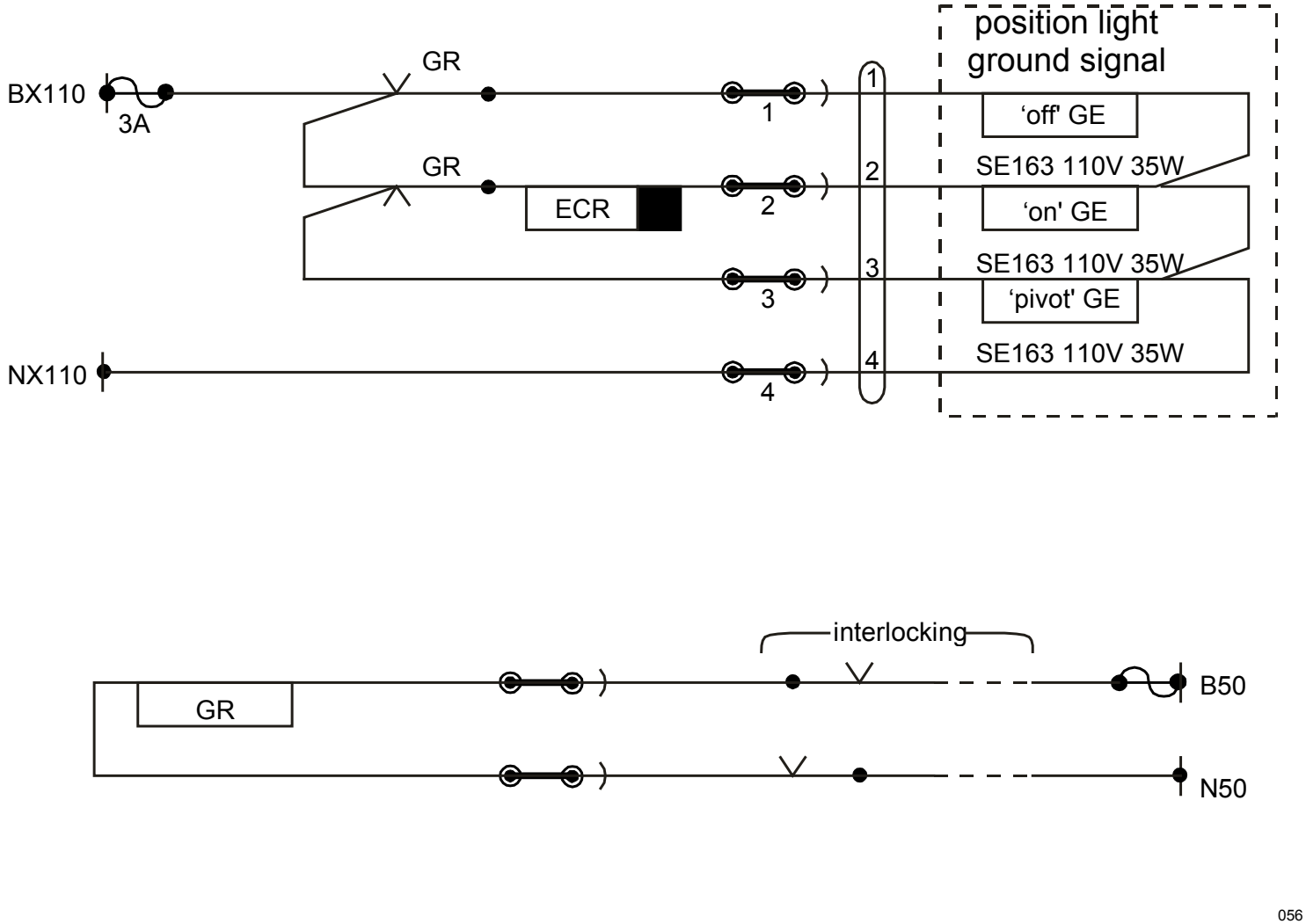


Figure 56

Lamp proving rarely used and usually confined to limit of shunt indication

For most applications the lamp proving relay ECR is not required and is omitted.

In the special case where the unit is used to mark the 'limit of shunt', the 'off' light position is blanked off, the 'on' and 'pivot' lights are both red and the ECR relay is used to prove the presence of current in both of them.

6.3 Operation of Position Light Subsidiary Signal

When used in association with a main aspect colour light signal, the red light and the back light of the position light signal are blanked off.

These units are used where the approach speed is not more than 15 m.p.h. and can be used as either

- a shunt signal
- a call on signal

The signal, when illuminated, has two white lights at 45°. Subject to certain conditions it allows the driver of a train to pass the main aspect at red. See the Rule Book Section C for details on what these are.

Circuit for a Position Light Subsidiary Signal

Figure 57 shows the circuit for a position light subsidiary signal. Before the subsidiary signal can show 'off' the main signal must be proved to be showing a red aspect and any associated route indication must also be proved.

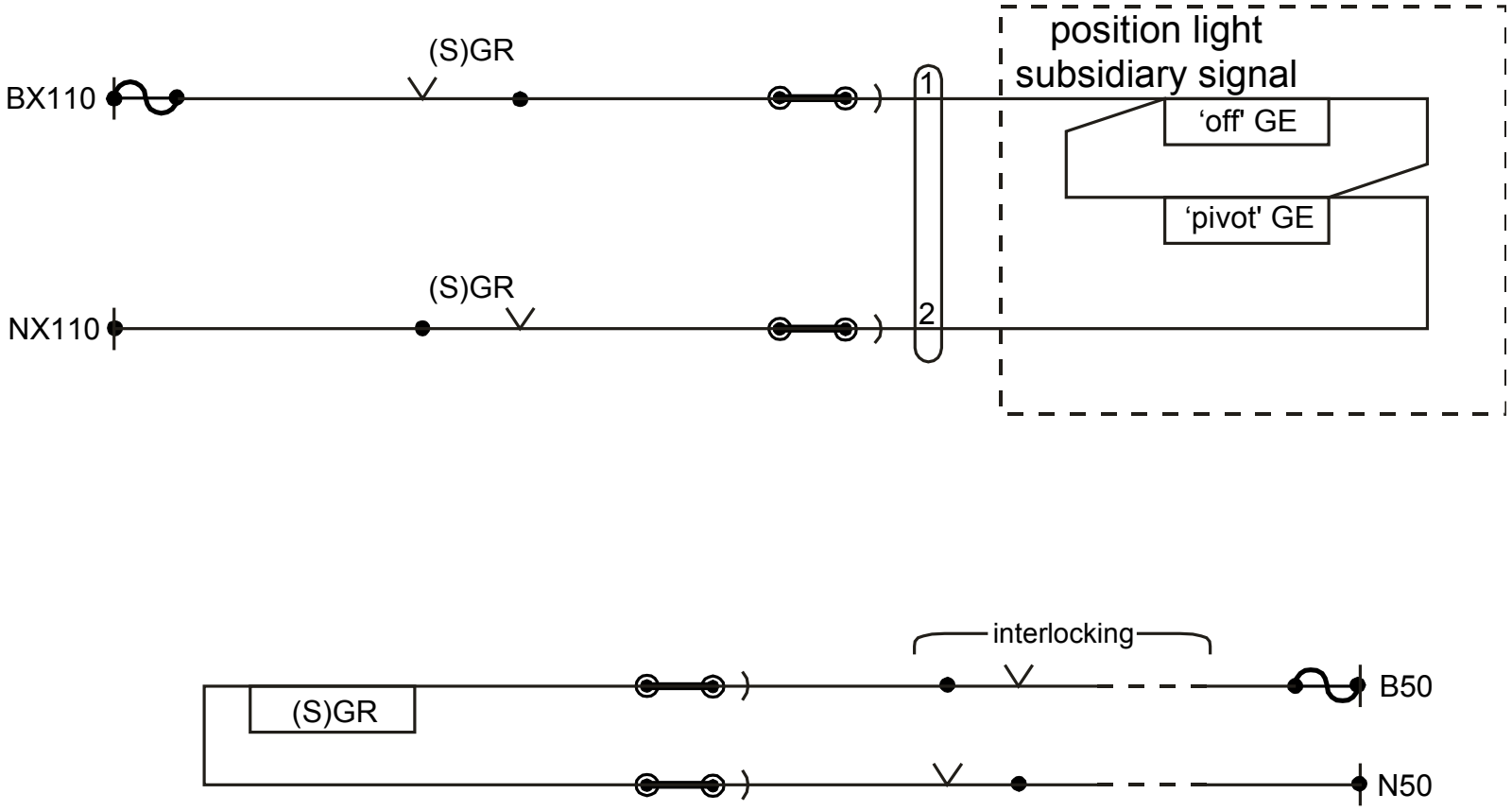


Figure 57

6.4 Section Summary

Section Six has looked at

- position light signal
- position light ground signal

Now complete Self-Assessment Question 8 before proceeding to Section Seven.

Self-Assessment Question 8

When a position light subsidiary signal displays two white lights at 45° what other signals must be alight?

Now check your answer with the one given in the Answers Section. Before moving on to Section Seven make sure that you fully understand all the information given in Section Six. If there are any areas which you do not understand, re-read the relevant parts of the text.

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7 Automatic Warning Systems (AWS)

7.1 Introduction

The purpose of an automatic warning system is to provide visual and audible indications to the driver as to the state of the signal aspect which

- is just being arrived at, or
- has just been passed

The automatic warning system is advisory and leaves the regulation of the train speed to the driver in most cases, but the system will apply the brakes automatically if the driver fails to react within four seconds when approaching a restrictive aspect.

Standard Arrangement of AWS Track Equipment

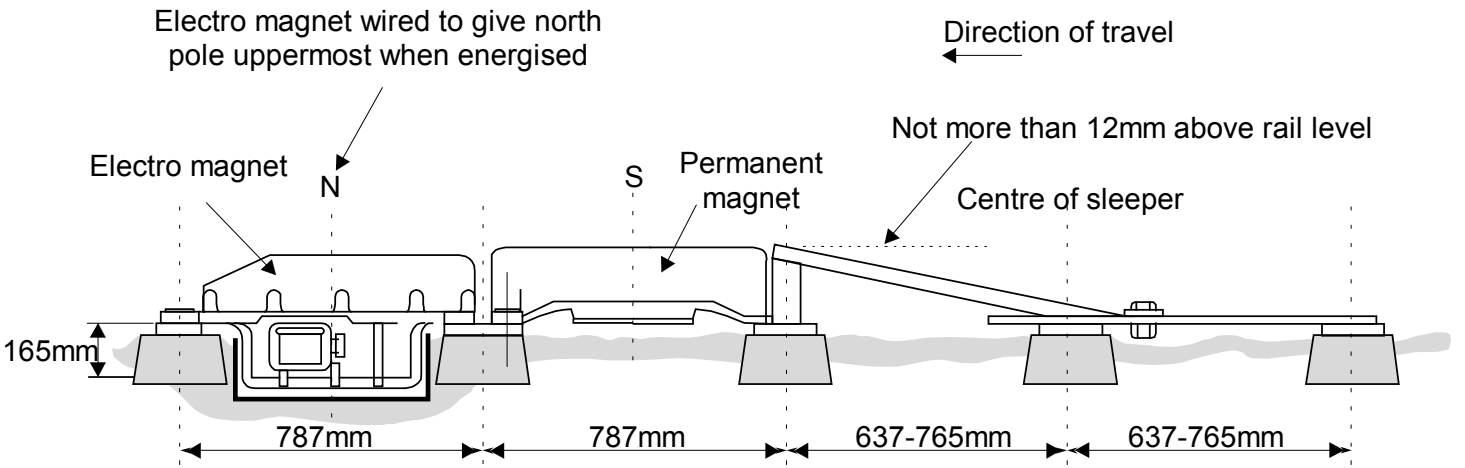


Figure 58

In areas where AWS is fitted the track equipment is normally provided at every semaphore distant signal and at every main colour light signal. The standard position for the track equipment is in the centre of the four foot at approximately 183 metres on the approach side of the signal.

The standard arrangement of the track equipment consists of two magnets and a ramp. In the normal direction of travel a train passes over

- the ramp, used to protect the equipment from hanging couplings etc.
- then the permanent magnet which has its south pole uppermost and gives a warning indication
- finally the electro-magnet which has its north pole uppermost, which cancels the warning and gives a clear indication only if the signal is indicating line clear.

The magnets are mounted at rail level plus or minus 12 mm.

Permanent Warnings

In some circumstances, such as the approach to a fixed distant signal or the warning board for a permanent or temporary speed restriction, there is a requirement to always warn the train driver to reduce the train speed. In these cases the ramp and the permanent magnet are fitted, but the electro-magnet is omitted. For temporary speed restrictions a portable permanent magnet unit is often employed.

7.2 Operation of the Automatic Warning System

The automatic warning system reflects the two states of the signal which will be either

- stop, caution or preliminary caution
- clear and proceed

In normal operation if the signal associated with the automatic warning system is displaying a restrictive aspect, namely

- red
- yellow, or a semaphore distant signal indicating caution
- double yellow
- the driver is given a caution indication.

If the signal displays green or is a semaphore distant signal at off then an indication is given to the driver that the signal is indicating line clear.

Restrictive Signal

The caution indication consists of a continuous note from a horn. If the driver fails to acknowledge the warning the brakes will be applied automatically. If the driver acknowledges the caution indication a visual indication is then presented to him as a reminder that he has taken control of the train. This remains until the train passes over another automatic warning system.

Clear Indication

The clear indication consists of a one second ring on a trembler bell or an electronic warbler giving a similar sound.

Visual Indication

Sunburst Indicator

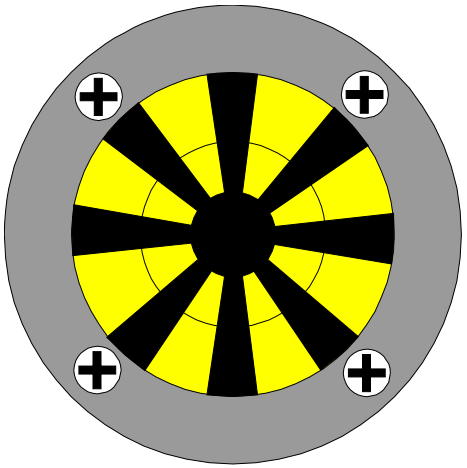


Figure 59

The visual indication takes the form of a sunburst indicator positioned on the driver's panel. This can be either

- yellow and black segments
- all black

When the warning horn has been cancelled the driver must act in accordance with the signal aspect exhibited.

The visual indicator is in no way connected with the signal aspects. Its purpose is to provide a reminder that the driver has acknowledged the warning horn and is in sole charge of the brake.

7.3 Components of the Automatic Warning System

Ramp

The ramp protects the equipment in the normal direction of traffic. When bi-directional traffic applies two ramps are provided.

Permanent Magnet

This has a south pole uppermost.

Electro-Magnet

When the electro-magnet inductor is energised it has a north pole uppermost.

Figure 60 is a typical example of an inductor circuit for an automatic warning system.

Circuit for AWS Inductor

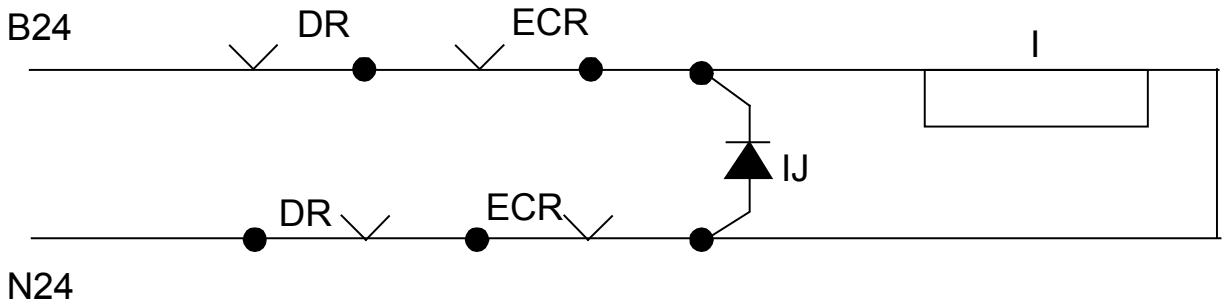


Figure 60

Figure 60 shows that to energise the inductor coil the green signal control relay (DR) and lamp proving relay (ECR) must be energised, proving that the signal is either

- displaying a green indication or
- a semaphore distant signal is off

There is a rectifier or diode (IJ) wired in parallel with the inductor, this is to prevent arcing across the relay contacts caused by the high back e.m.f. when breaking the inductive circuit.

Bi-directional lines and Suppressed Magnet

On bi-directional lines there may be a need to suppress the permanent magnet. This comprises an electro-magnet surrounding the permanent magnet. When the electro magnet is energised it cancels out the magnetic field of the permanent magnet, so in effect no magnet exists. Also on bi-directional lines where two opposite signals are sited so that one AWS system can serve them both, the arrangement can be.

- ramp
- suppressed magnet
- electro-magnet
- suppressed magnet
- ramp

Acknowledgement of Warning

On receiving the warning horn the driver should immediately acknowledge the warning by operating the re-set plunger. This will stop the horn sounding, prevent the automatic application of the brake and cause the indicator, normally showing all black, to display yellow spokes. This visual indicator will continue to show the yellow spokes until the locomotive or multiple-unit train passes over the next permanent magnet, when it will be automatically restored to all black.

7.4 Section Summary

This section has looked at the

- components of the Automatic Warning System
- operation of the Automatic Warning System

Now complete Self-Assessment Question 9.

Self-Assessment Question 9

What indication is given to the train driver by the AWS when the signal is green?

Now check your answer with the one given in the Answers Section. Make sure that you fully understand all the information given in Section Seven. If there are any areas which you do not understand, re-read the relevant parts of the text.

8 Train Protection Warning System (TPWS)

8.1 Introduction

In spite of the installation of AWS (see section 7) over most of the UK's main line railways, there has been a gradual increase in the number of signals passed at danger (SPADs) in recent years and some serious collisions as a result. In an attempt to reduce these, the UK Railway Inspectorate has made a number of suggestions for reducing the number of SPADs. One of these is the Train Protection and Warning System or TPWS.

The idea behind TPWS is that, if a train approaches a stop signal showing a danger aspect at too high a speed to enable it to stop at the signal, it will be forced to stop, regardless of any action (or inaction) by the driver.

The Train Protection and Warning System (TPWS) is a combination of the older AWS and additional automatic train protection system consisting of a train stop loop and an overspeed sensor.

8.2 How Does TPWS Work?

On the approach to each TPWS fitted signal, an overspeed sensor (OSS) and a train stop sensor (TSS) are provided.

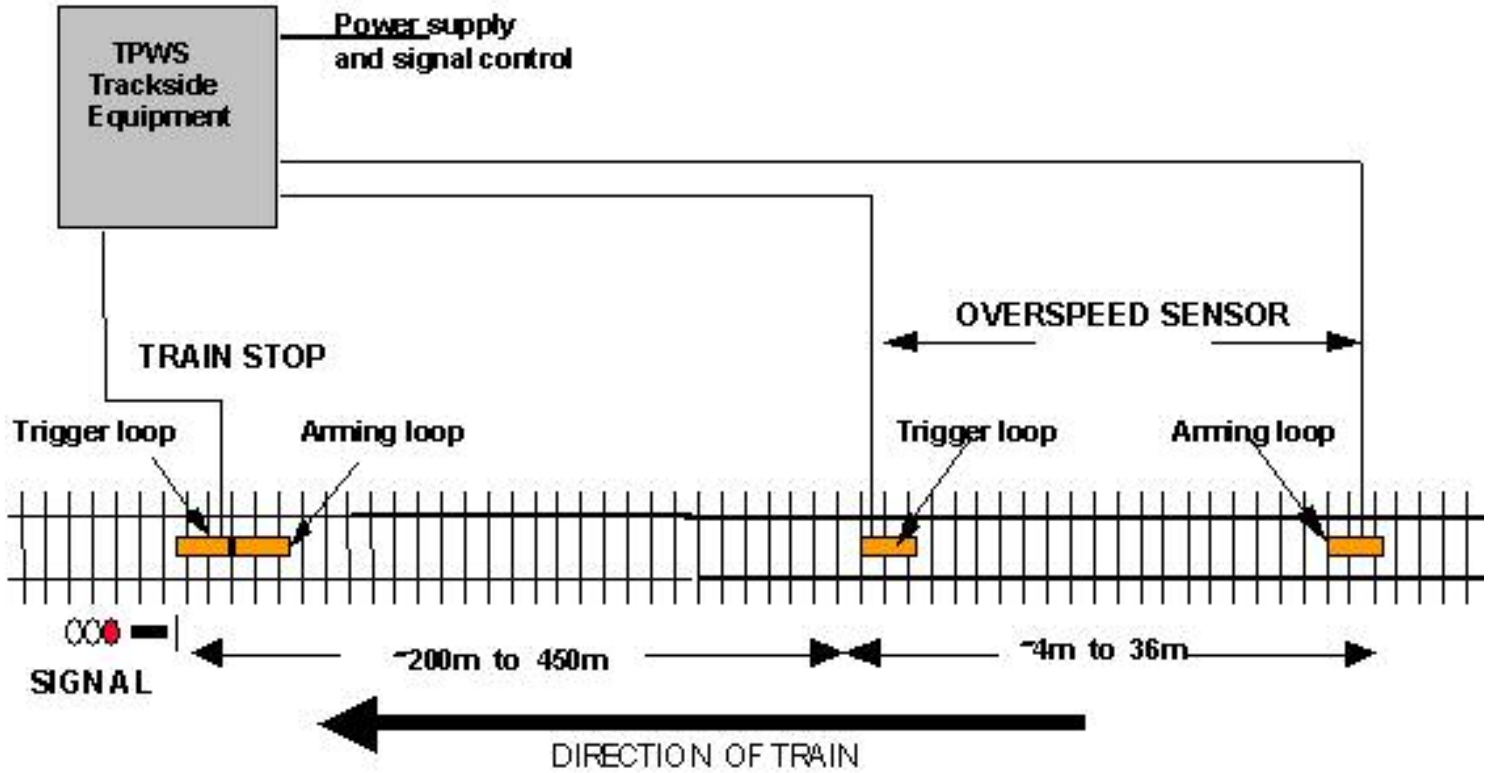
Both the OSS and TSS have two loops, an arming loop and a trigger loop. TPWS will only be energised if the signal is at red.

If the TPWS loop is energised it emits a radio signal, when a train passes over the arming loop this starts a timer on the train. If the train passes over the trigger loop with the timer still running then an emergency brake application is made. The driver cannot override the brake application for a period of one minute.

The timer on the train is usually set at one second for passenger trains; freight locomotives have a longer setting as freight trains generally run at slower speeds and require a longer distance to stop.

As the timer on the train is fixed, the only way to vary the trip speeds is to alter the distance between the arming and trigger loops. As an instant trip is required at the train stop, the loops are positioned next to each other.

A typical outline of a TPWS track layout for a selected signal junction



8.3 Why TPWS?

The prime advantage of TPWS as an installation programme is that it can be added to both track and trains independently. Thus it can be implemented quickly. Its introduction aims to provide a substantial reduction in the number of SPAD related accidents and so significantly mitigate deaths and serious injuries.

The Railway Safety Regulations 1999 require that a train protection system be operational on or before 1st January 2004 on:

- All trains on passenger carrying lines
- All signals protecting potential conflicts with passenger trains
- All buffer stops approaches in passenger platforms
- All approaches to permanent speed restrictions (PSR) where a speed of >60 mph is reduced by over one third.

8.4 How Effective is TPWS?

TPWS will stop a passenger train with emergency braking within a 200 yard overlap up to about 75mph.

TPWS is a non-intrusive system. If the train is driven within permitted operating speeds and is brought to a stop before any red signal, the system will not intervene. TPWS operates as a last resort intervention system; it uses the fact that the overlap exists.

Stopping a train is not as simple as stopping a car. Each carriage of a train weighs about the same as a lorry and the grip between the wheels and the rail is about the same as that for a car on an icy road. Thus trains do not stop quickly. For this reason there is a safety overlap beyond a red signal as a train approaches so that if a minor mistake is made the train will remain safe. This overlap (or safe overrun distance) is typically 200 yards

TPWS only measures the train speed at one location as the train approaches a signal at red. The overspeed equipment needs to be placed such that it will not normally interfere with a driver operating his train correctly. Avoiding unnecessary intervention and technical limitations means that the overspeed equipment needs to be not more than 500 metres before the signal. A modern train will stop within the safety margin from 75 mph given these constraints.

At speeds above 75 mph the system still works and causes the brakes to be applied but the train may stop beyond the safety overlap. Whether there is an accident will depend on the track layout and what other trains are in the area. Thus Network Rail have not claimed it will prevent such accidents. In some cases the accident will be avoided and in others the speed of collision will be reduced. However the majority of SPADs are at lower speeds and the system will have a significant effect on these.

8.5 What happens if TPWS fails on the infrastructure?

On the trackside, the equipment will provide an automatic fault indication to the signaller who can then notify the maintainers to fix the fault. The safety of the railway is maintained by returning the signal behind to danger. When the signal no longer needs to be red because the line ahead is clear the signal can be cleared and the signal behind will then clear.

8.6 TPWS Equipment

The TPWS equipment is designed to interface with the existing on-board wiring of trains so that the additional equipment can be fitted quickly.

Bogie Mounted Antenna



A TPWS indicator unit is fitted to the train equipment along with a TPWS antenna

Driving Cab Layout



TPWS loop (one half of an OSS)



Pandrol clip holds the Rail to the Sleeper fixing for support beam



TPWS SIM Modules



Signalling Interface Module, Over Speed Sensor module and Train Stop Module.

The modules are colour coded to indicate both function and frequency of operation and hence the direction of travel for which they should be used.

8.7 Section Summary

This section has looked at the

- components of the Train Protection Warning System
- operation of the Train Protection Warning System

Now complete Self-Assessment Question 10.

Self-Assessment Question 10

How long is the timer usually set for on the train for passenger trains?

Now check your answer with the one given in the Answers Section. Make sure that you fully understand all the information given in Section Seven. If there are any areas which you do not understand, re-read the relevant parts of the text.

9 Module Summary

This Module has looked at

- semaphore signals
- colour light signals
- route and junction indicators
- position light signals
- automatic warning system
- train protection warning system

Before attempting the Self Test Questions which follow make sure that you fully understand what you have read. Take your time and check your answers to the Self-Assessment Questions again. When you feel confident begin the Self Test Questions.

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10 Self Test

Attempt the following Self Test Questions. Take your time and think about the answers carefully. When you have finished the Self Test check your answers with those given in Answer Section. If you experienced any difficulties return to the relevant part of the text and re-read the material. Do not move onto the next module until you feel confident in your understanding of this work.

Self Test Question 1

What type of shunt signal would be used on the layout below and why?

Location of Shunt Signal

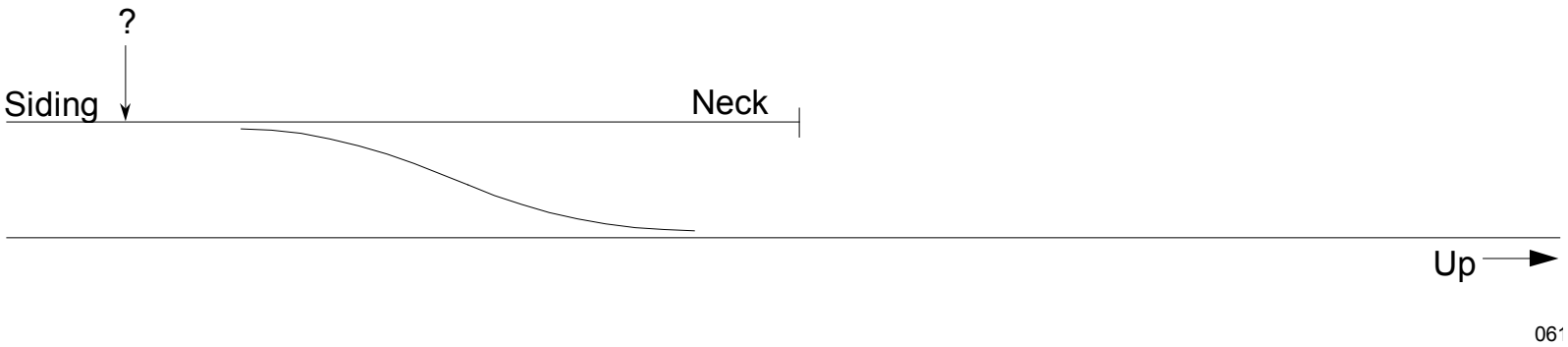


Figure 61

Self Test Question 2

Draw the stop arm of a mechanical stop signal.

Self Test Question 3

How many lenses are used in each aspect of a multi aspect signal and which one is coloured?

Self Test Question 4

Name the three types of route and junction indicator currently in use.

Self Test Question 5

What are the two applications for the position light signal.

Self Test Question 6

If a signal is at double yellow, what indication is received by the driver from the AWS system?

Self Test Question 7

What happens if TPWS fails on the infrastructure?

Self Test Question 8

Briefly describe how TPWS Work?

11 Answers

This section provides the answers to both the

- Self-Assessment Questions
- Self Test Questions

Answers to Self-Assessment Questions

Self-Assessment Question 1

The night time aspects of a mechanical stop signal are

- when the signal is on a red light is shown and the driver must stop
- when the signal is off a green light is shown and the driver can proceed to the next stop signal

Self-Assessment Question 2

The information given by a semaphore distant signal when it is off is that all the relevant stop signals controlled by the same signal box are off.

Self-Assessment Question 3

The mechanical shunt signal with a red bar cannot be passed until it is off.

The yellow band shunt signal can be passed when it is on for certain movements.

Self-Assessment Question 4

A subsidiary signal at off and revealing a letter C is indicating a call-on. See Rule Book C4.2.2

Self-Assessment Question 5

Current will flow through the front contact of HR through the back contact of the DR to the signal head transformer for the yellow aspect and double yellow aspects.

Self-Assessment Question 6

The auxiliary lamp voltage is adjusted with link 1 broken to disable the main filament and by moving the appropriate slider on resistor GEO. Then link 1 is restored and the main lamp voltage is adjusted by moving the other slider on resistor GEO.

Whilst the main filament is intact the relay ECR (M) is energised by the current flowing through the main filament. The front contact of ECR (M) energises the auxiliary filament through the high resistance coil of ECR (A). The resulting small current does not light the auxiliary filament but does energise ECR(A).

If the main filament blows the relay ECR (M) is de-energised and its back contact energises the auxiliary filament through the low resistance coil of ECR (A) The resulting current lights the auxiliary filament and energises ECR (A).

Provided the auxiliary filament is intact ECR (A) is energised when the main filament is on or off. If the auxiliary filament blows, ECR (A) is de-energised when the main filament is on or off.

Self-Assessment Question 7

Three lights need to be alight in a position light junction indicator before the main aspect will clear.

Self-Assessment Question 8

The other signals that will be alight are the main red aspect, plus some form of route information if that is appropriate.

Self-Assessment Question 9

When the signal is at green, the driver will receive a bell sound.

Self-Assessment Question 10

The timer on the train is usually set at one second for passenger trains.

Answers to Self Test Questions

Self Test Question 1

The shunt signal required is a position light ground signal with a yellow light or a mechanical shunt signal with a yellow band across, so that a train may proceed up the shunt neck without having the signal cleared first.

Self Test Question 2

Semaphore Stop Signal Arm

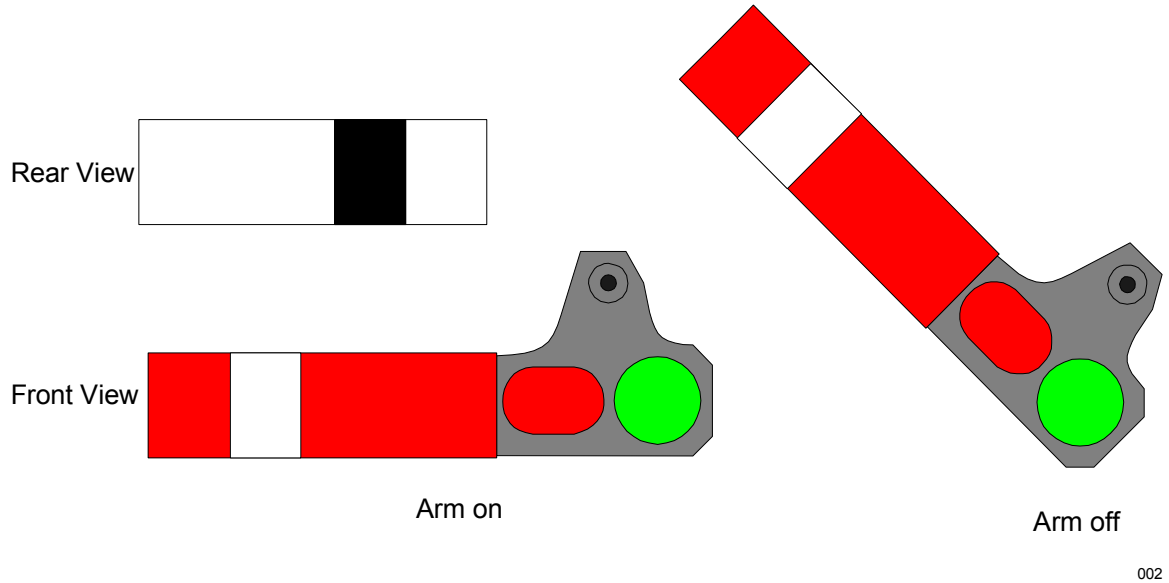


Figure 62

Self Test Question 3

Two lenses are used and the inner lens is coloured.

Self Test Question 4

The three types of route and junction indicator are

- position light junction
- theatre type
- stencil

Self Test Question 5

The position light signal is used as

- a shunt signal
- a subsidiary signal

Self Test Question 6

The AWS system will give the driver a horn indication when a signal is at double yellow.

Self Test Question 7

If TPWS fails on the infrastructure?

The equipment will provide an automatic fault indication to the signaller who can then notify the maintainers to fix the fault. The safety of the railway is maintained by returning the signal behind to danger. When the signal no longer needs to be red because the line ahead is clear the signal can be cleared and the signal behind will then clear.

Self Test Question 8

On the approach to each TPWS fitted signal, an overspeed sensor (OSS) and a train stop sensor (TSS) are provided.

Both the OSS and TSS have two loops, an arming loop and a trigger loop. TPWS will only be energised if the signal is at red.

If the TPWS loop is energised it emits a radio signal, when a train passes over the arming loop this starts a timer on the train. If the train passes over the trigger loop with the timer still running then an emergency brake application is made. The driver cannot override the brake application for a period of one minute.

11 Glossary

Within the railway system there are a number of regional variations and individual conventions which mean that alternative terms may be used for

- equipment
- operations
- features

Module Five has used terms which normally correspond with those found in the Signalling Installation Handbook. However, it may be useful for you to familiarise yourself with the alternative terminology which you may encounter in your work. This glossary provides a number of variations which may occur.

Short range outer lens	-	spread light lens
Hot strip	-	prism segment

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